

Commonsense Psychology and Language

By the time we have become fluent speakers of our native languages, we have learned to use thousands of words and phrases to refer to mental states and processes of ourselves and others. This richness in vocabulary parallels the complexity of the commonsense psychological model that defines the deep lexical semantics of these linguistic expressions. By studying the richness of psychological language, we learn about the various facets of the knowledge representation challenge, and its overall breadth of scope.

In this chapter, we attempt to catalog the full breadth of English words and phrases related to commonsense psychology. We organize this vocabulary into twenty-nine representational areas, and group words and phrases into lists of entries that we consider to be conceptually synonymous. Undoubtedly we have missed quite a few expressions, and we have not attempted to list every synonym or morphological variation. Instead, our goal is specifically about breadth of coverage and granularity of conceptual distinctions. In creating these word lists, we are defining the target of our knowledge representation efforts. Our task is to develop a set of formal theories of commonsense psychology in which all of these words have a referent, and in which the theories themselves are rich enough to distinguish between each enumerated concept. We concede that there are subtle differences between words and phrases that we have grouped into a single enumerated concept. However, we believe that the concept categories capture much of the meaning of each group of expressions, and give us a good starting point for future efforts to articulate (formally) subtle differences in meaning.

In offering this (long) catalog here, we hope to achieve several goals. First and foremost, we can think of no better way to show readers the enormous breadth of commonsense psychology in terms that are already well understood. Second, we see these lists as a non-technical index to the topics that are formalized in Part III of this book. Where readers are interested only in certain topics of commonsense psychology, one need only to find the relevant concepts in these lists and consult the corresponding logical formalization. Third, these lists invite alternative formalizations of commonsense psychology, providing a starting point for those that want to pursue radically different approaches with identical conceptual breadth. Fourth, it serves as a new lexical resource for future natural language processing systems, especially those that aim to interpret language using commonsense knowledge.

1. Eventualities and Their Structure

Because commonsense psychology deals with the things people think about and because people think about states and events in the world, we need to maximize the convenience of talking about such things. This is done by making states and events first-class individuals in the logic. That is, states and events are things in the world. They can be referred to by constants and variables in the logic. This enables us to use

them as arguments in predications and therefore represent the properties of states and events. We “reify” states and events, from the Latin word “re(s)” for “thing”; we take them to be things. We will use the term “eventuality” to cover both states and events, after Bach (1981).

Eventualities may be possible or actual.

2. Substitution

We will frequently have occasion to talk about eventualities that exhibit the same pattern, are similar, or are tokens of the same type. Thus, we need predicates for describing this similarity in structure.

3. Typical Elements

Among the things we can think about are both specific eventualities, like “Fido is barking,” and general or abstract types of eventualities, like “Dogs bark.” We do not want to treat these as radically different kinds of entities. We would like both, at some level, to be treated simply as eventualities that can be the content of thoughts.

4. Instances

If x is a typical element of a set and x_1 is a member of the set, we can say that x_1 is an instance of x . We would like to extend this notion to eventualities.

5. Composite Entities

A composite entity is a thing composed of other things. It is one of the most basic concepts in a knowledge base of commonsense knowledge. It is hard to think of anything that is not a composite entity, and much of our everyday vocabulary is for talking about composite entities.

Under the concept of “composite entity” we mean to include complex physical objects, such as a door, a cup, a telephone, a chair, and an automobile; living beings, such as a tree, a honey bee, and a person; complex events, such as a hike, the process of erosion, and a concert; and complex information structures, such as an equation, a sentence, a theory, and a schedule. Some entities, such as books, have both physical and informational components. In this theory of composite entities, we do not make any distinctions among the types of components an entity might have. From the standpoint of the theory of composite entities, the physical-abstract distinction is of no interest.

A composite entity is characterized by a set of components, a set of properties, and a set of relations.

The Figure–Ground Relation

The figure–ground relationship is of fundamental importance in language and cognition. We encode this with the predication $(at\ x\ y\ s)$, saying that a figure x is at a point y in the ground s .

Patterns And Their Instances

A pattern is a composite entity that contains type elements among its components.

6. Scales

A scale is a set of entities with a partial ordering among them. Thus, it is a composite entity where the components are the members of the set and one of the relations is the partial ordering. It is of course possible for particular scales to have other properties and other relations.

Scale-To-Scale Functions

We can extend the notion of function to scales by saying that for a scale to be the domain or range of a function is for its set of components to be the domain or range.

Qualitative Structure On Scales

There is a range of structures we can impose on scales. These map complex scales into simpler scales. For example, in much work in qualitative physics the actual measurement of some parameter may be anything on the real line, but this is mapped into one of three values: positive, zero, and negative. Where the parameter is vertical velocity, this means we are interested only in whether something is going up, staying at the same elevation, or going down.

An even simpler scale is one with only two points, where one is less than the other. Any predication defines such a scale, where the two points are the predication being false and the predication being true, and where the former is less than the latter. When we axiomatize arithmetic in Chapter 13, it will also be a scale, and of course it imposes a very common kind of structure on other scales, namely, measurements of quantities. In addition, we will define a “half-order of magnitude” structure for scales that is frequently useful in making approximate judgments.

7. Arithmetic

Integers

In a practical reasoning system, one would not try to solve arithmetic problems by automatic theorem-proving from basic axioms. One would simply use the procedural attachment capabilities of the theorem-prover and utilize the computer’s much more efficient arithmetic operations. Nevertheless, for conceptual completeness, we will axiomatize basic arithmetic here, following the standard Peano axiomatization, in terms of a successor predicate.

Rational Numbers

Fractions will be characterized by a nonnegative integer (the numerator) and a positive integer (the denominator). The predication (fraction f a b) says that f is the fraction a/b . Such fractions exist.

Measures And Proprtions

Sets of rational numbers, and hence sets of nonnegative integers, are very important examples of scales. We will focus on sets in which 0 is the smallest element. If e is the lt relation between x and y and $s1$ is a set of numbers containing 0 but

no smaller number, then there is a nonnegative numeric scale s with s_1 as its set and e as its partial ordering.

8. Change of State

It is hard to imagine a universe without change. It is even harder to imagine cognition in such a universe. Change of state is very nearly the most basic concept human beings are equipped with. We could not survive very long without recognizing the changes in our environment. Our very sense of time arises from our awareness of change.

9. Causality

The account of causality we use here is that of Hobbs (2005b). This distinguishes between the monotonic, precise notion of “causal complex” and the nonmonotonic, defeasible notion of “cause.” The former gives us mathematical rigor; the latter is more useful for everyday reasoning and can be characterized in terms of the former. We begin with an abbreviated account of these concepts.

When we flip a switch to turn on a light, we say that flipping the switch caused the light to turn on. But for this to happen, many other factors had to be in place. The bulb had to be intact, the switch had to be connected to the bulb, the power had to be on in the city, and so on. We will use the predicate `cause` for flipping the switch, and introduce the predicate `causalComplex` to refer to the set of all the states and events that have to hold or happen for the effect to happen. Thus, the states of the bulb, the wiring, and the power supply would all be in the causal complex.

Agents And Agenthood

There are some entities in the world that are viewed as being capable of initiating a causal chain. This is a scientifically inaccurate view; when a dog stands up and walks across a room, there are events in its brain that caused it to do so. But the idea pervades commonsense reasoning. We say that the dog’s walking was caused by the dog, and don’t necessarily expect to find anterior causes.

We will call such entities agents. People are the prime examples of agents, but the class also includes robots and other intelligent software, higher animals, organizations, and a variety of fictional entities such as gods, ghosts, and goblins. Frequently, when we use a person metaphor for some other type of entity, agenthood is what motivates us.

Ability

The concept of “ability” is difficult to characterize. It is closely related to possibility. We will see in Chapter 20 that possibility is characterized with respect to a set of constraints. Something is possible if those constraints do not rule it out.

Ability is also with respect to an implicit set of constraints. Suppose Joan is sleeping when we ask, “Is Joan able to play tennis?” The answer is clearly no if we include her sleeping as one of the constraints. If we don’t, it may well be yes.

But when we speak of an agent’s ability to do something, we generally remove from consideration eventualities that are beyond the agent’s control. For example, is Joan able to play tennis if all the tennis courts within reach are already occupied. In that case, it would not be possible for her to play tennis, but she is still able to play tennis. A person’s ability to perform an action is normally viewed as the action being possible provided all the eventualities not under his or her control go the right way.

Executability

In computational settings, it is relatively easy to say what “executable” means. An action is executable if it is directly implemented in the underlying software and its preconditions are satisfied. But more generally, the notion of “executable” is a matter of perspective. From one perspective, one can view driving home from work as an executable action. From a finer-grained perspective, one has to decompose this into actions such as drive a block and turn right. At an even finer granularity, we can take as the executable actions such things as maintaining a certain pressure on the accelerator and turning the steering wheel by a certain amount. There is no limit in principle to how fine-grained the decomposition can be, although when talking about human plans, we are generally satisfied calling something executable if it is an automatic action that does not require conscious thought.

We can posit a notion of “directly causes” (dcause) as a relation between an agent or event and an event, which is true when there are no intermediate, mediating events.

Difficulty

If an action is difficult for an agent, it is because there are states and events, that is, eventualities, that tend to cause the action not to happen. We will attempt to capture this intuition in a predicate called `difficult` that takes an action and its agent as arguments.

10. Time

Temporal Ordering And Interval Relations

Instants are at least partially ordered by a before relation. The before relation will also be extended to intervals, temporal sequences, and eventualities, so the constraints on its arguments are as follows:

Durations

There are two ways we could introduce measures of the duration of intervals.

The first is to map the time scale directly to the real numbers, and define duration in terms of the difference between the numbers that the beginning and end of the interval are mapped to.

The second is to introduce a relation of `sameDuration` between intervals and to stipulate that certain intervals are standard durations, called “temporal units.” Then any hour-long interval has the same duration as the standard hour. Durations of arbitrary intervals are then defined in terms of concatenations of unit-duration intervals. We will take the second approach since it is probably closer to the way the notion of duration arose in evolution and arises in individual development.

We will assume that people can make the judgment that two intervals are of the same duration. The predicate `sameDuration` basically applies to intervals, but trivially every instant is of the same duration, namely, zero. So the constraints on the argument of `sameDuration` are that its arguments are temporal entities.

Periodicity

A temporal sequence is periodic if there is a constant duration between any two successive temporal entities in the temporal sequence.

Rates And Frequency

We would like to define a notion of “rate” that will cover the following cases, among others:

He was driving 60 miles per hour.

We have 3 meetings every 4 1/2 months.

I see 4 students a day.

We discuss three topics every meeting.

In the most primitive case, there is a set S of events. In our four examples, the event types are driving a mile, having a meeting, seeing a student, and discussing a topic.

There is also a set T of time intervals that may or may not be all of equal duration. In the first three examples, the durations are one hour, 4 1/2 months, and one day. In the fourth example, the lengths of the meetings may differ. There is a function mapping the intervals into a subset of S, namely, those events that occur during that interval. If the concept of “rate” applies truly, the cardinalities of all these subsets are the same. This analysis provides us with all we need to define the primitive case. The concept of “rate” will be extended beyond this in the text that follows, so here we use *if* rather than *iff*.

11. Space

Space And Spatial Analogies

Many “top-level” ontologies begin with a distinction between physical objects and abstract entities. By contrast, we have made it through thirteen background theories without ever mentioning the distinction. The reason for this is that the core of language doesn’t seem to care much about this distinction. We can be in a building, and we can be in politics and in trouble. We can move a chair from the desk to the door, and we can move the debate from politics to religion and move money from one bank account to another. Ontologies that begin with this distinction, or similar ones like Cyc’s tangible–intangible distinction (Lenat and Guha, 1990), fail to capture important generalizations in language and as a result very nearly make themselves irrelevant in linguistic applications at the outset.

It has frequently been observed that we understand many abstract domains by analogy with spatial relations (e.g., Vico, 1744; Richards, 1937). A relatively recent rediscovery of this old truth was by Lakoff and Johnson (2008). We operate in space from the moment we are born and thus build up very rich spatial models. By setting up a mapping between a new domain and space, we are able to commandeer these rich models for the new domain and think about it in more complex ways.

This would seem to indicate that the very first theory one should develop in an enterprise such as ours is a very rich theory of space, and then set up some mechanism for analogical reasoning. But what does analogical reasoning involve? It is a matter of identifying common properties of the things being compared, reasoning about them in the familiar domain, and then transferring the results to the new domain. But what is typically transferred from space to new domains in spatial analogies is not just any property. Very rarely, for example, do we transfer the properties of hardness or color or precise distance. The common properties that are transferred are usually

topological properties.

In these background theories we have identified some of the most important underlying properties of the spatial domain that are most frequently utilized in analogies – such things as complex structure, scalar concepts, change of state, and so on. We have constructed theories of these abstract properties. The content of these theories is very similar to proposals in work in cognitive linguistics (e.g., Croft and Cruse, 2004) on the “image schemas” that seem to underlie much of our thought.

If one were inclined to make innateness arguments, one position would be that we are born with an instinctive ability to operate in spatial environments. We begin to use this immediately when we are born, and when we encounter abstract domains, we tap into its rich models. The alternative, more in line with our development here, is that we are born with at least a predisposition toward instinctive abstract patterns – composite entities, scales, change, and so on – which we first apply in making sense of our spatial environment, and then apply to other, more abstract domains as we encounter them. This has the advantage over the first position that it is specific about exactly what properties of space might be in our innate repertoire. For example, the scalar notions of “closer” and “farther” are in it; exact measures of distance are not. A nicely paradoxical coda for summing up this position is that we understand space by means of a spatial metaphor.

It is also a curious and perhaps surprising fact about our efforts to construct the cognitive theories of Part III that very few purely spatial concepts turned out to be necessary. We occasionally need to make reference to physical objects, to one physical object being at another, and to a physical object being near another. And that is all.

Spatial Systems And Distance

We will use the predicate *physobj* – (*physobj* *x*) – to say that something is a physical object. We will not explicate this notion, because it would take us deeply into commonsense physics rather than commonsense psychology. We would have to talk about properties of color, weight, and malleability. Instead we will simply stipulate that something is a physical object, and not attempt to constrain the possible interpretations of the predicate *physobj* by means of further axioms.

Location

In Section 10.3 we introduced the figure–ground relation (*at* *x* *y* *s*), meaning that external entity *x* is at component *y* of composite entity *s*. A spatial system is a possible ground by virtue of the fact that all its components are physical objects. That is the common property they have that makes a unitary interpretation of the *at* relation possible (cf. Axiom 10.14).

12. Persons

Finally we arrive at people. The theories of Part III are intended to some extent to apply to other kinds of agents than just people, such as robots and organizations, and some aspects of the cognitive theories, such as goals, plans, and beliefs, we would expect to find in any cognitive agent in some form. But many aspects are idiosyncratic to people – accidents of evolution, in a sense. For example, there is probably no reason a robot or an organization should be thought of as having emotions. In Part III, when we are talking about aspects of cognition that apply to all cognitive agents, we will call the agent simply an “agent.” When we are talking about particularities of people,

we will condition the axioms on the relevant arguments being persons.
A person is a kind of agent.

13. Modality

Eventualities exist in a Platonic universe of possible (and impossible) individuals – entities, states, and events. If they happen to actually occur in the real world, that is one of their properties, and we express it with the predicate *Rexist*. Real existence can be thought of as one possible mode of existence. There are others. An eventuality could be part of someone’s beliefs but not occur in the real world. It could exist in someone’s imagination. It could exist in some fictional universe. It could be merely possible rather than real. It could be likely. It could be unlikely, or not real, or impossible. An especially important modality is “happening at a particular time” or *atTime*. We have gone to some effort in this book to simplify the logical formulas as much as possible. Using unprimed predicates when possible and primed predicates only when necessary has been one way of doing this. Another issue we debated was whether to include time arguments in every predication, and we decided against it in order to keep the notation relatively simple. But that raises the question: Just what is the relation among unprimed predicates, *Rexist*, *atTime*, and temporal arguments?

Positive Modalities

Of the various modalities, the most important are what can be called “positive modalities.” Generalizations can be captured at this level, and we could do so by introducing the predicate *PosMod* as a cover term for all positive modalities. *PosMod* takes two arguments. Although we will only use *PosMod* applied to eventualities, there is no reason it cannot be applied to other entities as well. Its second argument will be a predicate that labels the modality.

Possibility And Necessity

Possibility is with respect to a set of constraints. For example, is it possible to put an X in the central square in tic-tac-toe if your opponent has already put an O there? No, if you accept the rules of the game. But if the only constraints you accept are the laws of physics, then, yes, you can. It is possible.

Likelihood, Or Qualitative Probability

Possibility is one common judgment we make about eventualities in situations of uncertainty. Another is probability or likelihood. But we have a terminological problem in developing a qualitative theory of probability, namely, what to call it. “Qualitative probability” is too cumbersome. “Probability” by itself means something else very precise to a large number of people, the mathematical theory of probability, and it would be too confusing to use the term for something different. The term “likelihood” is also a very precise technical term – the probability not of the output of a model but of the parameters of the model given its output, for example, the probability that a coin is fair given the results of a series of tosses – although this technical term has the advantage of being less widely known. The ad hoc nominalizations “probable-ness” and “likely-ness” are barbarisms. The words “chance” and “chances” seem to have more to do with opportunities and good outcomes. “Odds” is a technical, mathematical term that is very widely known. We will make the best of a bad set of options and use the term “likelihood,” but it should be understood that it is not the same as what is meant in statistics by the term. We mean not the cleaned-up, mathematical version, but the qualitative, vague judgments we make in

everyday life, as when we say it is likely to rain without having any real mathematical basis for the judgment. We might be able to make a rough guess, such as “a 60% chance of rain,” but we’d have no basis for deciding between a 60% chance and a 59% chance.

When ordinary people talk about the likelihood of some event occurring, they do not have in mind one of the set of real numbers between 0 and 1. For us, likelihoods will be elements of a scale, but rather than being totally ordered, they will be only partially ordered. We might wonder what the likelihood is of our favorite television show presenting a new episode this week, and we might wonder what the likelihood is of the Alpha Centauri system having more than ten planets, but we’ll rarely have to decide which is more likely, and if we did have to make that decision, we’d be hard-pressed to find a basis for it. The Likelihood Scale will have a top and a bottom, where the top means the event is certain to happen and the bottom means it is certain not to happen, but the top and bottom will not be identified with the real numbers 1 and 0. Since likelihoods are not numbers, we cannot apply arithmetic operations like addition and multiplication to them. We can only state some fairly weak ordering relations. Nevertheless, all the axioms we state will be true in the mathematical theory of probability. Probability will be one model of our qualitative theory, just not the only one. The predicate likelihood will express a relation between these likelihoods and eventualities. In addition, like possibility, likelihood is with respect to an implicit set of constraints that in a sense defines the sample space. Making the constraints an argument allows us to relate likelihood to possibility and to entailment.

14. Knowledge Management

Knowledge Management concerns the basic reasoning and inference processes that lie at the core of the commonsense view of human psychology. The conceptual scope of this area, evident in the following linguistic examples, does not veer far from the philosophical concerns in the classical Platonic dialogue *Theaetetus*, which put forth a definition of knowledge as justified true belief, and the field of epistemology that follows from this work. Also included are process-oriented concepts within this representational area, such as the mental actions of adding and removing beliefs and assumptions, checking for inferences, the realization of consequences, and the affirmation of beliefs.

Objects Of Belief

A fundamental concept in a theory of knowledge management is “belief.” Traditionally, theories of belief concern propositions, agents, certain relations among propositions, and certain relations between agents and propositions. However, in Chapter 5, we rejected the distinction between propositions and eventualities as at best superfluous. At worst, propositions do not divide the world up in a sufficiently fine-grained fashion to be the objects of belief.

Belief

There are three primary properties of belief that must be captured, all of them defeasible. First, we can do logic inside belief contexts. Defeasibly, we believe the consequences of our beliefs, though there can be notable lapses. Second, perceiving is, at least defeasibly, believing. Third, our beliefs influence our actions.

Belief Revision

Changing beliefs, or belief revision, is a topic on which there has been a great deal of research in artificial intelligence. We will examine one important issue, without, however, trying to address all of the extensive literature.

The conditions under which these knowledge-management operations are possible may be quite complex. For example, we probably don't want to say that someone who knows no physics is capable of adding a belief concerning string theory. Similarly, if someone is asleep or merely fatigued, we may not want to say they have the ability to add a belief.

Individual people have greater or lesser tendencies to add new beliefs and perform other knowledge-management actions. For example, some people are more gullible than others, believing the statements of others without further justification; others are more skeptical, and more likely to doubt the statements of others. More on this in the next section.

In most work on belief revision (Peppas, 2008), one is assumed to have a consistent set of beliefs. A new fact, generally certain, is learned that is inconsistent with the belief set. The belief set then has to be restored to consistency. A number of methods and criteria have been proposed for this operation, but usually the intuition is that we try to make as few changes as possible in our existing belief set.

One usually assumes logical omniscience; the belief set is taken to be deductively closed. This is not realistic; we don't know all the logical consequences of what we know. But the operations for restoring consistency in that framework when an inconsistency exists are going to be the same as the operations in our framework when an inconsistency is discovered. Thus, the assumption of logical omniscience does not make that work irrelevant.

It is also usually assumed that the new fact that triggers the inconsistency is something we are certain of, and it is thus the prior belief set that must be altered. This is not always the case. If we see a colleague flying down the hallway, we may decide simply to disbelieve what we saw, rather than make the massive changes in our commonsense knowledge that would be required to remove the inconsistency. But when we discover that we "believe" both P and $\neg P$, we have to withdraw belief in one of the two, and the process for doing so will be the same. So we can frame the problem as one of restoring a belief set to consistency by withdrawing belief in $\neg P$ and whatever would imply it.

Most often when we revise our beliefs, we do so not by withdrawing assent to some belief but by complicating or otherwise modifying the beliefs. Where the belief set consists mostly of implicative rules, we will generally try to find rules where we can tighten the antecedent or loosen the consequent to escape the contradiction. For example, suppose we believe that whales live in water and that things that live in water are fish. We then learn that whales are not fish, and we restore our belief set to consistency by tightening the antecedent of the second rule to something like "lives in water and has gills." A child with limited experience with dogs may believe all dogs are brown.

Degrees Of Belief

There are many issues where we believe neither a proposition nor its negation with absolute certainty. Rather we sort of believe it, or sort of don't believe it, or we view it as a toss-up. In a treatment of commonsense psychology, we need not just a theory of belief but also a theory of "graded" belief. We need the notion of believing something to some degree.

Assuming

Agents are able to assume propositions. Generally the reason for doing this is that reasoning can be done within assumption. We assume a proposition, do some reasoning, and determine what else would be true.

One function of assuming is hypothesis testing. We assume a hypothesis and examine the consequences. If a consequence is false, we know the hypothesis is false. If all the consequences we draw are true, it will strengthen our belief in the hypothesis.

Indirect proof is a variety of hypothesis testing. We assume a proposition is false and we prove that a falsehood results, thereby demonstrating that the proposition is true.

A proof by cases involves successively assuming the full range of possible values for some variable and proving the proposition in question for each of them. When we combine that with the fact that the cases are exhaustive, we have a proof.

We make assumptions also to suppress certain reasoning processes and their consequent actions. Suppose a student offers a teacher a bribe for a grade, and the teacher replies, "I'm going to assume you didn't say that." The alternative is to follow the consequences of the offer, which may result in more serious difficulties than either of them wants.

When we assume something for the sake of the discussion, we are pretending that the proposition is true in order for our partner in the conversation to express consequences that depend on it, without committing ourselves to belief in the proposition. We are able to reason about that assumption. For example, we can judge whether an argument is coherent, given the assumption.

Inference

When an agent a infers an eventuality e from a set s of eventualities, there is a causal relation from a 's belief in s to a 's conscious belief in e .

Justification

Justification is closely related to inference. A set s of premises justifies something e for an agent a if a believes e as a result of inferring e from s .

Knowledge

Beliefs can be true or false. We can define "true" and "false" in terms of real existence,

Intelligence

The intelligence of agents is extremely difficult to characterize. It has a great many components. One of these components is the ability to draw inferences, and here we introduce the intelligence scale and constrain its interpretation in terms of this ability. Other factors in determining intelligence are the ability to think of analogies, creativity or the ability to come up with new ideas, and the speed of drawing inferences. But we will not attempt to characterize these factors or to present a general theory of intelligence.

We will not attempt to give necessary and sufficient conditions for the partial ordering moreIntelligent. But it is consistent with the ability to draw inferences.

If Pat is able to draw all the correct inferences from a set of premises that Chris can draw and then some, then defeasibly Pat is more intelligent than Chris. Indeed, this is precisely how we often compare the intelligence of two people – seeing how well they solve the same problem set. The rule is defeasible because Chris may outperform Pat

on other criteria for intelligence.

15. Similarity Comparisons

Similarity Comparisons concerns the comparison of things, real or imagined, based on their representations. Along with basic notions of similarity and difference, this area includes other mental processes that are based on similarity judgments, specifically the ability to recognize patterns and draw analogies between otherwise dissimilar things.

Similarity

People are very good at determining when and in what ways two things are similar. In this chapter we introduce several notions of similarity, of increasing complexity.

Two things are similar if they have a property in common. A dialog expanding on similarity might go as follows:

16. Memory

In the commonsense view, memory is a metaphorical container for thoughts. Storing and retrieving from memory involve the transfer of thoughts in and out of this store to one's focus of attention. Included are concepts involved in deliberative memory retrieval as well as for passive retrieval, where a memory cue triggers an involuntary reminding.

Storing And Retrieving

Recall from Section 21.6 that we said an agent has a mind, and that a mind has at least two components, a memory and a focus of attention. Intuitively, the focus of attention is where conscious thought takes place, and the memory is where concepts, or memories, are stored. As in Chapter 21, we will allow objects of thought to be concepts or the entities and eventualities the concepts are of. That is, when an entity or eventuality is treated as an object of thought.

Accessibility

Concepts in memory have an “accessibility” which is an element in a partial ordering.

Associations And Causing To Remember

One concept can remind an agent of another concept. This occurs when the first concept being in focus causes the second to be remembered.

17. Envisioning

What are you thinking about? It is a common question that people can easily answer, and one that we will want to pose to robots in the future. Just as with people, we expect that robots should respond with one of handful of types of reports. They might be thinking about the past and how things got to be the way they are now. They might be focused on the present moment and the current state of the world. They might be wondering about the future and what is going to happen next. Perhaps instead they are daydreaming, imagining fictional places and people that have no temporal or causal connection to the world that they are experiencing. These processes all have

a few important commonalities. They each involve the manipulation of conceptual models of the world, real or fictional. They each are concerned with events and states within these models that have a temporal and/or spatial extent, and the temporal and causal relationships between them. Each involves the consideration of the possible, freely considering counterfactual events and states as easily as those that are experienced in reality.

The phrase “thinking about” is probably the most commonsense way of referring to the suite of reasoning processes that share these commonalities, but in this book we instead use the term “envisioning.” The concepts in this area characterize the infrastructure of causal reasoning, including branching causal timelines of various likelihoods that organize states and events into a system of relationships. Included as well in this area are concepts that pertain to the management of a person’s current model of the world, such as advancing this model through the current moment, sending the present to the past and the future to the present, and validating or invalidating the expected model.

Thinking Of

In this chapter we address the very difficult issue of what it is to think about something. There are many varieties of thinking about something, but they all involve having concepts in the focus of one’s attention, that is, being conscious of them. To think of some concept, entity, or eventuality is to have it in the focus of attention and for that to cause other concepts, entities, and eventualities that it is associated with to come to be in the focus of attention as well. We will refer to the state of having something in the focus of attention as “thinking of” it.

Causal Systems

In Chapter 15 we defined a causal complex for an effect as all of the eventualities that have to obtain for the effect to occur. An eventuality is “causally involved” in an effect if it is in a causal complex for that event. An eventuality “causes” an effect if it is the critical element in a causal complex for the effect in the sense that all the other elements of the causal complex normally obtain, but it does not. Of the three notions, “causally involved” is the weakest.

In this chapter we wish to define a “causal system” as the sort of thing people have in mind when they think causally. It will be weaker than a causal complex, because people never have in mind all the relevant eventualities. It will be based on the “causally involved” relation. Essentially we want a directed graph where the nodes are eventualities and the arcs are “causally involved” relations.

In addition, people often have disjunctive branches in mind when they think causally. One of two eventualities might result, or one of two eventualities might have been the cause. These disjunctions constitute branches in the directed graph. So we first define a “branch” and then define causal system arcs to include branches.

Contiguous Causal Systems

In causal thinking we move incrementally from one envisioned causal system to another, in a sense, by growing or pruning one node at a time. For “causally involved” arcs, we can subsume adding and deleting causes and effects under one relation between two causal systems. The relation `oneArcDiff` says that the two causal systems `c` and `c0` differ by only one arc, the arc `r` from `e1` to `e2`. Causal system `c` will be the larger one, the one that is obtained by adding arc `r` to `c0`. If `e1` is already in

c_0 and e_2 is not, then this relation captures the operation of adding or deleting an effect. If e_1 is not in c_0 and e_2 is, then the relation amounts to an operation to add or delete a cause. But the relation `oneArcDiff` is more general in that it also covers the case in which both e_1 and e_2 are in c_0 and the case where neither e_1 nor e_2 is in c_0 .

Envisioned Causal Systems

When we are thinking causally, we have a contiguous succession of causal systems in mind. Here we first define what it is to envision a causal system (ecs). We then define a contiguous sequence of them (ecsSeq).

The easiest way to define it would be to say that it is a causal system that the agent has in focus, or is thinking of. But we will give a somewhat weaker definition. The agent a is (consciously) thinking of every eventuality in the causal system. But a may or may not be thinking of all the causal relations. They may be merely believed and not in focus, or not even necessarily accessible. For example, when we rapidly assess a situation by intuition, we are using causal knowledge we may not even be aware of. On the other hand, the agent may be consciously thinking of and considering the consequences of a causal rule that is not believed. So we will stipulate that the arcs in the causal system are either believed or in focus.

Envisionment And Belief

In Chapter 21 we said that believing a set of eventualities is believing their conjunction. To believe a causal system is to believe the conjunction of the set of eventualities and the set of causallyInvolved relations that characterize the causal system.

Other Varieties Of Thinking

Our account of envisioning is an attempt to capture what it is to think causally. But there are many other varieties of thinking. We will not go into any of these in detail, but it may be useful to sketch out how one would approach characterizing them.

In every case, we would characterize them as a sequence of sets of eventualities and entities that are in the focus of attention, together with other properties, for example, whether or not they are believed or whether or not they are unpleasant or by what kind of relations they are linked to each other.

There are many varieties of thinking that can be viewed as specializations of envisioned causal system sequences, and some of these are discussed in subsequent chapters, such as planning, explanation, and monitoring the execution of one's plans. Also similar to envisioned causal systems is mathematical thinking, where we are tracing out implicational networks, trying to determine whether eventualities we introduce into the system are necessary or sufficient, and trying to determine everything that must be included. Another example is where we are trying to categorize something we perceive – is that a cat or a raccoon? – by running over diagnostic properties in our mind. This is often important for predicting behavior. A kind of meta-causal reasoning occurs when we ponder whether some aspect of our belief system needs revision; we consider our beliefs and the good and bad consequences of having those beliefs. In all these cases, we are trying to keep an ordered succession of ideas in focus. In defining envisioned causal system sequences, we proposed one common kind of ordering.

Another kind of ordering is similarity. Often episodes of thought linked by similarity are only mental play, as when we see the shape of a dog in a cloud. But similar things often behave similarly, so such similarity reasoning is often functional. We often are able to surmise how to behave in a new situation because it reminds us of

a situation we are familiar with.

Another variety of thinking occurs when we are very much engaged in the moment, and mostly what we are consciously thinking is the perceptible situation immediately before us and what our next action should be. For example, an orienteer racing through the very uneven terrain of a forest must be very alert to obstacles and ready to respond to anything that might suddenly appear. Someone performing a new and complex dance is in a similar situation, as is a boxer.

By contrast, fantasizing is disengaged from the current immediate situation. Many of one's beliefs are suspended, while many beliefs are maintained as background assumptions. For example, one might counterfactually imagine having been a child prodigy while holding fixed one's beliefs in what is interesting. A formal characterization would need to deal with both what beliefs are held fixed and what beliefs are up for grabs. Dreaming is a similarly not apparently functional mode of thinking. One would have to characterize the kind of awareness that occurs in sleep, for example, as having much the feel of perception, but with only a narrow window on the world and coupled with an inability to act.

There are also compulsive dysfunctional episodes of thought, for example, in mental fussing where we think of the same problems or irritations over and over again with no hint of problem-solving activity. Another example is when we cannot get the image of a gruesome accident out of our mind. These kinds of thinking seem to involve a high degree of repetition and no apparent progress toward the achievement of our goals.

We leave the characterization of these and other varieties of thinking to future work.

18. Explanation

People want to know why. How did the world end up this way? What caused this to happen? The answers to these questions are explanations, the product of the mental process of explanation. Explanation is the side of envisionment that extends models of interconnected events backwards in time, to what came before to cause other events that are considered. The concepts below outline a multistage explanation process that begins with an attempt to explain a mystery, and is followed by the generation and evaluation of candidate explanations that are either adopted or rejected.

Explanations And Mysteries

In Chapter 24 on Envisioning we defined what it is for an agent a to explain an eventuality e with eventuality $e1$. The agent believes that $e1$ caused e . Our notion of explanation pertains only to those eventualities e the agent believes actually happened. In this chapter we discuss explanations more fully, including what it means for one explanation to be better than another, an idealized account of the process of explanation, and how explanations can fail. Abilities and possibilities are constrained by the agent's current world understanding.

The Explanation Process

An idealized picture of the process by which an agent comes up with an explanation is as follows: First the agent recognizes the need to explain an eventuality. Candidate explanations are then generated. The candidates are evaluated to see which one is the best. The inferior candidate explanations are rejected, and the winning candidate explanation is adopted and becomes part of the agent's beliefs. In this section, we

define a predicate for each of these steps, in part to show that we have developed the vocabulary required for characterizing this process.

Explanation Failures

The notions of trying, success, and failure are explicated in Chapter 28 on Goals. Suffice it to say here that to try to do something is for one to have it as a goal and for that to cause one to carry out some actions in a causal complex that would lead to it. One succeeds when the goal is achieved, and one fails when it is not. An explanation failure is a failure to explain.

19. Managing Expectations

What is going to happen next? Expectations are envisionments of causal consequences, specifically those that are imagined to follow in the future from the current world state. Expectation is the side of envisionment that extends interconnected models of states and events forward in time, toward the temporal and causal consequences of other imagined states. As the world reveals itself, these expectations may be confirmed or violated, the latter of which is the basis for the state of surprise.

In the chapter on Envisioning, we defined an “envisioned causal system” (ecs) as a set s of eventualities entertained by an agent a with a particular set of causal connections among the eventualities. We defined the “current world understanding” (cwu) to be the ecs s that an agent a believes to be currently the case at time t_0 – (cwu0 s a t_0). We defined what it is for two causal systems to be contiguous – (contigCS s_1 s_2). In terms of that, we defined an “envisioned causal system sequence” (ecsSeq) to be a sequence of contiguous ecs’s. This provides us with the background we need to formalize expectations and their justifications.

To expect eventuality e to happen or hold at some time t_1 in the future generally results from having an envisioned causal system sequence, one of whose elements is the current world understanding and one of whose downstream elements contains the eventuality e . By “downstream” we mean that the time t_1 of e must be after the “Now” time t_0 that anchors the expectation. That is, e occurs in the future with respect to the expectation. However, we first define expectation only in terms of belief and then use envisioned causal systems to justify the expectation.

20. Other-Agent Reasoning

Other-Agent Reasoning concerns the imagining of the mental states of other people, as well as introspectively considering one’s own mental state. By including these concepts here, we are arguing that a theory of thinking about mental states is itself part of commonsense psychology, that is, that how people think they think includes beliefs about how people think about how people think! This idea is perhaps best understood in consideration of the representational concept of a psychological theory, included below, for which the theory formalized in this book is an instance. Despite the title of this representational area, it is also meant to include the mental processes of self-reflection and introspection, where people think about their own mental processes from the privileged perspective.

People have the ability to reason about the mental state of other people. It presents no new difficulties for us to represent this, because mental events are just a particular kind of event and our notion of causality covers causal relations between mental events and other events. We have treated reasoning as envisioning within a causal

system. Thus we can define what it is to reason about someone's mental state in terms of this.

21. Goals

People want a lot of things: a loving partner, a fulfilling career, parents who are well cared for, an attractive physique, a creative outlet, and material possessions. These are the sorts of goals that people target in their pursuit of happiness. The central importance of goals in people's lives is reflected in the richness of language about goals. The list of goal concepts below are abstract goal categories, rather than specific goals that people choose to pursue. A specific instance of a goal may change its class over time. For example, the preservation goal of caring for the health of one's parents may first only be an envisioned future goal when they are young and in good health, a shared goal among siblings when their health needs attention, an auxiliary goal when they are involved in one's plans in some way, and a violated preservation goal when their health turns bad.

Goals, Subgoals, And Plans

Human beings are intentional agents. We have goals, we develop plans for achieving these goals, and we execute the plans. We monitor the executions to see if things are turning out the way we anticipated, and when they don't, we modify our plans and execute the new plans. A theory of goals and planning can be applied not just to people but also to other entities that can be conceived of as agents, such as organizations and complex artifacts. In fact, it is a not uncommon cognitive move among people to attribute agency even to natural phenomena such as volcanos and hurricanes. Anything that seems to exploit and manipulate the causal structure of the world as a means toward some end can be viewed as a planning mechanism.

The Content Of Goals

There are a number of types of goals, characterized by their content and the agents who have them, that figure large in strategies, and in this section we define a number of these classifications to show that we have built up machinery adequate for describing them.

Any eventuality can be a goal. A particularly important kind of goal is a goal of knowing something, because most actions have knowledge prerequisites.

Goals And Multiple Agents

Individual persons are not the only kind of intentional agent. Sufficiently complex devices of various sorts can be viewed as intelligent agents. So can collectives of agents, whether collectives entirely of people, or assemblages of people and devices. These collectives can have goals. For example, General Motors has the goal of selling cars. They can devise plans for achieving goals. General Motors' plan involves manufacturing and marketing cars. These plans must bottom out in the actions of individual persons or devices, or in things that will be true at the appropriate time anyway.

For example, we can have a plan to get your car started by pushing it a short distance to the top of a hill and then letting it pick up speed on the downhill side until it is fast enough that you can pop the clutch. This plan bottoms out in my individual action of pushing on the back of the car and your individual action of pushing on the frame of the open left door. Of course these actions have to be synchronized, but these are properties of the individual actions. The plan also bottoms out in the

event of the car rolling down the hill from the top. This is something that will happen anyway at the appropriate time, and doesn't have to be carried out by any member of the collective.

A shared or collaborative goal is a goal in which the agent having the goal is a collective. Moreover, the members of the collective mutually believe the collective has the goal.

Trying, Succeeding, And Failing

When we try to bring about some goal, we devise at least a partial plan to achieve it, including subgoals of the original goal which are actions on our part, and we execute some of those subgoals. Moreover, our executing those actions is a direct result of our having those actions as subgoals. We can take this as a definition of "trying."

Functionality

Complex artifacts are generally constructed for some purpose. Some real or hypothetical agent has a goal in mind, and the artifact achieves that goal. Cars, for example, have at least the purpose of moving us from place to place. The structure of complex artifacts typically reflects a plan to achieve the goal, where the various components are involved in some subgoal. For example, the steering wheel of a car is involved in the subgoal of having the car go in particular directions. We will call such a subgoal the functionality of the component. Organizations and their components can be analyzed similarly.

In fact natural objects can too, if we associate a hypothetical agent having as a goal the normal behavior the natural object tends to engage in. We can stipulate that the "goal" associated with a tree is to grow and reproduce, and we can analyze the structure of the tree as an instantiation of a plan to achieve that goal. We can then talk about the function of the various parts of the tree. We can even view volcanos, for example, as composite entities that have the associated "goal" of erupting, and talk about the functions of its parts to this end.

In general, almost any composite entity can be associated with a goal by some hypothetical agent, and where components are causally involved with the behavior of the whole, we can view the relation between an action by the component and the behavior of the whole as a subgoal relation. We can then define the functionality of the component as that subgoal relation. (We use the elevated term "functionality" rather than "function" because we have already used the latter to refer to mathematical functions.)

Plans to achieve goals are just a way of exploiting the causal structure of the world to achieve some end, so "goal talk" can be imported into any domain where the manipulation of causal structure is involved.

A composite entity that goes through changes can be viewed as having as an associated goal some element or subset of those changes. We will define that "goal" as the functionality of the whole. We call this predicate *functionality0* because it is absolute functionality, not the functionality of a component relative to the behavior of the whole.

Value, Cost, And Importance

The concepts of value and cost, in modern society, are often, if not usually understood in terms of money. This has the advantage of turning what is inherently a partial ordering in the commonsense understanding into a total ordering, allowing any two things to be compared. We will not take that approach here. Instead we will try to characterize the more basic concepts of value and cost in terms of the partial order-

ing. Cost and value are scalar notions, and for all scalar notions we need to introduce a partial ordering relation – moreValuable and moreCostly. As with many common-sense concepts, we cannot define these relations precisely, but we can constrain their interpretations by means of axioms relating them to theories we have developed independently, in this case the theory of goals. Then we can define a valueScale and a costScale in terms of the partial ordering. The predicates valuable and costly identify the Hi region of the scale, the value of something is its location on the valueScale, and the cost of something is its place on the costScale. An interpretation of cost and value in terms of money will be one possible interpretation consistent with our axioms, but not the only one.

The key idea we develop here is that achieving a subgoal is not as valuable as achieving its supergoal, and sacrificing a subgoal is not as costly as sacrificing its supergoal. This is consistent with the monetary view of value and cost because generally money enables goals to be achieved, and more money means more goals can be achieved.

Importance is less often monetized, although winning \$10,000 in the lottery is a more important event in your life than winning \$10. But importance is closely related to value and cost. The more valuable something is, the more important its acquisition. The more costly a sacrifice, the more important. Value is a positive notion about what is gained; cost is a negative notion about what is lost. In a sense, importance is a neutral notion that generalizes over these two. Valuable acquisitions and costly sacrifices are both important.

Here we will first characterize these concepts for eventualities in terms of their effects on goals. Then we will extend the concepts to entities in general, as anything can have a value, a cost, and an importance.

A concept, entity or eventuality is more or less valuable, costly, or important to an agent depending it's relation to the agent's goals. Thus the three predicates have the agent as an argument. What is important to one agent may be of no consequence to another. Moreover, the entities being compared are distinct.

22. Goal Themes

To capture a commonsense intuition, Schank and Abelson (1977) introduce the term “goal theme,” which “contains the background information upon which we base our predictions that an individual will have a certain goal.” In their formulation, the attribution of a given goal theme to a person provides a causal account for other goals that would otherwise be senseless if viewed in isolation. Three broad classes of goal themes were described by Schank and Abelson, which can be referred to as role, relationship, and life themes. We include the concept “thrive,” the causal antecedent of all of a person's goals.

Thriving

It is formally convenient to assume that agents have one plan that they are always developing, executing, monitoring, and revising, and that that plan is in the service of a single goal. We will call this goal “Thriving.”

Pleasure And Pain

Pleasure is sometimes taken as the motive for all action. We do things because we believe they will lead to pleasurable feelings. We will not take that approach, because we are characterizing intentional action as arising out of agents' goals and the plans they develop to achieve these goals. But there is indeed a close relationship between

pleasure and goals. If something is pleasurable, we are likely to pursue it as a goal. In fact, this is probably its primary property. A Martian or a robot who did not experience pleasure could go a long way in understanding the role of pleasure in people's lives just by knowing this rule. In terms of evolution, we can see the utility of a mechanism that would impel us toward actions that would generally be good for us, but at the same time can be overridden when we know better, for example, when we refrain from eating a decadent dessert. Like many mechanisms developed in evolution, pleasure is rough and ready, working most of the time, but subject to error.

Short-Term Versus Long-Term Goals

A very common conflict when we are trying to decide what to do is between satisfying short-term goals and pursuing long-term goals. In Aesop's fable of the ant and the grasshopper, the grasshopper pursues the short-term goal of enjoying summer, while the ant pursues the long-term goal of surviving the winter.

The terms "short" and "long" are relative to each other. In absolute terms the relevant delays can be minutes – "Don't fill up on bread before the main course comes" – to years – should you get a good-paying job now or should you go to medical school? The only constraint on the times is that one is before the other.

The goals have to be conflicting; otherwise we'd just satisfy the short-term goal now and the long-term goal a little later. Eating the bread now will spoil your appetite for the main course, and it's nearly impossible to have a full-time job and succeed in medical school.

Finally, the long-term goal has to be more valuable to the agent than the short-term goal. Otherwise we would always simply pursue the short-term goal.

We will abbreviate this situation with the unpronounceable predicate *stvslt*. Its arguments are an agent *a*, two desired eventualities *e1* and *e2*, and two times *t1* and *t2*. The constraints on the arguments are given in the following axiom.

Goal Themes

In general our knowledge is tagged by who else believes it. Academics generally assume that their fellow academics don't believe in parapsychology or flying saucers. Americans believe in democracy. A computer scientist will believe that other computer scientists, but not laymen, will know what the $P=NP$ problem is. Christians believe fellow Christians believe in the resurrection of Jesus, but don't believe that Jews and Muslims believe it. One of us was rather taken aback when he met a nun who didn't believe in God.

The mutual belief predication (*mb s e*), introduced in Chapter 21, says that a particular set of agents *s* mutually believes an eventuality or proposition *e*. That eventuality can be a general rule.

23. Threats And Threat Detection

The commonsense concept of a threat is somewhat remarkable in that it sits at the overlap between so many different ideas in commonsense psychology, particularly those related to one's goals, one's plans, and expectations about the way the world is going to be. Combined, these ideas provide a concise definition: a threat is an expectation that if something is not done, there is some chance that some goals will not be achieved. In everyday English, the term also often refers to the causal antecedents

Threat Situations

A threat situation is one in which the agent sees that things could turn out badly. To formalize this notion, we need to assume there is a causal system s that is envisioned by the agent a . The threat is somewhere downstream causally from some situation e_0 , perhaps the present situation in the agent's current world understanding, but perhaps not. Threats are generally not inevitable. Thus, there is in the causal system a branch e between e_1 and e_2 , where one of the alternatives, say e_1 , leads to one of a 's goals being negated and the other alternative does not lead to that goal being negated.

Threat Detection And Management

To detect a threat is to change into a state in which a situation s is envisioned as a threat situation. Threat detection is often thought of as a more active process. The agent goes through an envisioning process in which the final envisioned causal system is a threat situation. This may be real or hypothetical depending on whether the envisioned causal system is the current world understanding or not.

Seriousness

A threat can be more or less serious to an agent. Seriousness is a composite scale that depends on importance and likelihood. Of two equally likely threats, the more important one is more serious. Of two equally important threats, the more likely is more serious. This composition of scales is captured in the predicate `compositeScale` introduced in Chapter 12. The likelihood scale was introduced in Chapter 20. The importance scale was introduced in Chapter 28

24. Plans

Here the term “plan” is used in the same sense as in this quote from Daniel Burnham (1864–1912), the celebrated American architect and city planner, “Make no little plans. They have no magic to stir men's blood and probably will not themselves be realized” (Moore, 1921). Burnham's quote is emblematic of his design philosophy, as evidenced by his development of the influential Burnham Plan for the city of Chicago, but the quote also directly implies some of the characteristics of plans in general. First, plans are seen as something that are made by people, as Burnham advises the listener to do. Although Burnham's plans often were presented as physical objects (architectural drawings), it is natural as well to view plans as entirely mental constructs, the product of human reason. In the commonsense view, therefore, a plan is a thing that resides in one's head. Second, plans can be small or large. For Burnham, the size of a plan may have been directly correlated to the size of the building, neighborhood, or city that he was envisioning, but a plan's size may also be based on the quantity or severity of the steps involved, or simply the duration that it would take to execute. Third, one person's plan can be communicated to other people for appraisal. When Burnham says that small plans cannot “stir men's blood” he is noting that people can have the plans of other people in their head to appraise their value, without the necessity that those plans be adopted or pursued. The idea that someone could have in his or her head an “un-adopted plan” makes perfect sense in everyday discourse, but much of the theoretical work in artificial intelligence (AI) planning seems to struggle uncomfortably with this notion, leading to a terminology of desires, intentions and commitments that take on nonobvious meanings within the literature. Fourth, plans are either realized in the world or not. For Burnham's realized plans, the world is changed in readily apparent ways: old roads and buildings

are demolished, new ones take their place. Other plans may leave no obvious trace of their execution, such as a plan to have a casual conversation with a friend, or to relax at home on a Saturday afternoon. Despite the lack of obvious change to the world, the realization of these plans is dependent on the occurrence of real-world events.

Plans As Mental Entities

We saw in Chapter 28 on goals how plans result from the use of causal knowledge in trying to achieve goals. In this chapter we discuss plans as mental constructs, something that can be remembered, can be applied many times, may not be completely thought out, can be shared among agents, and so on.

To define this, we first need to define the reflexive transitive closure of the subgoal relation between two eventualities e_1 and e_2 . In addition, we want all of the eventualities to be in a specified set s_1 and all of the pairwise subgoal relations to be in a specified set s_2 . This will insure that the plan is connected.

The Planning Process

We think and we act. We often think without acting, and we often act without thinking. In commonsense psychology, these two capacities are viewed as distinct. We will do that here by dividing cognition crudely into a belief system and a planning module. A creature with only a belief system would do nothing, however rich its model of the world. A creature with only a planning module without beliefs about the state of the world would not survive long. We can view a belief system as a large network of interconnected beliefs, including beliefs about what causes or enables what, but with no impulse to action. The planning module has goals and uses causal beliefs to develop plans to achieve these goals. The belief system concerns what agents know; the planning module concerns what they try to accomplish. The first is about knowledge; the second is about action.

Thus, we can identify three realms in which causality figures. The first is the world with its causal complexes linking ensembles of events.

The second is an agent's belief system, including beliefs about what kinds of things cause what other kinds of things. Desires can be viewed as a particularly important kind of causal belief, a belief about what will contribute to one's thriving.

Desires are usually viewed as weaker than goals. A woman might desire a million dollars, but not work for it, whereas if she has the goal of acquiring a million dollars, it is reasonable to assume she is performing actions that lead in that direction. In a move that is certainly open to considerable debate, we will assign goals to the planning module and desires to the belief system. In a way, we will take "desire" to be what in the belief system corresponds to "goal" in the planning module. Desire will be modeled as a belief, possibly contradicting other beliefs, about what will cause one to thrive at a particular instant. Desires can thus be seen as a variety of beliefs about causes.

Strategies And Tactics

Causal knowledge can be used on an ad hoc basis to develop plans for specific goals. However, we also ruminate about how to achieve general classes of goals, developing general "strategies" for achieving them at some later time. Consider the strategy of securing customer loyalty by giving coupons (Gordon, 2004). The causal basis of this strategy is the following:

1. Giving a coupon to someone enables him or her to use it.
2. The use of a coupon causes the price of the item to be less.

3. The less the price of an item, the more a person prefers to buy the item at the business that sells it for that price.

4. People tend to do what they prefer to do.

We could further unpack the third rule in terms of a person's maximizing available resources. The broad-scale characterization of this strategy is the rule, "Secure customer loyalty by issuing coupons."

A strategy can be seen as a large-scale structure of causal knowledge that can in principle be unpacked into individual causal rules, and rests on these rules, but is generally used as a whole. Moreover, it is knowledge about types that can be instantiated in any number of tokens. For example, we can instantiate the above rule for any number of specific stores, kinds of coupons, and classes of customers.

A strategy like this is a plan, in the sense that it is a way of exploiting the causal structure of the world to achieve certain goals. The term "strategy," however, is usually reserved for very general classes of circumstances. We would not normally call something a strategy if it was a plan that could be employed only once and only with a specific individual. Not all of the entities involved in the strategy need be general. A man may have a strategy for persuading his wife, and only her, of something. But generally if we are to call this a strategy, then the class of situations in which it would be applied would be general

Executability And Complete Plans

The definition for plan allows plans to be incomplete. In fact, a single goal with no subgoals is a limiting case of a plan. So what makes a plan "complete"?

A plan is a way of manipulating the causal structure of the world to achieve one's goals. It must bottom out in events that are assured to occur at the proper time. There are three sources for this assurance:

1. The event is an action on the part of the agent that the agent is capable of carrying out at the required time.
2. The event will happen anyway at the required time, because of external causes.
3. The event is an action on the part of another agent, who is committed to performing the action at the required time.

For example, a trapeze artist has the plan to swing out on her trapeze, fly through the air, and be caught by her partner. She knows that she can swing out on her trapeze because this is an action that is executable by her. She knows that she will fly through the air; the laws of physics will take care of that. She knows her partner will catch her because he is committed to doing so. Thus, her plan is complete.

Types Of Plans

Since a plan can be viewed as a set of subgoals and subgoalOf relations among them, it can be viewed as a conjunction of these subgoals and relations, and hence as a logical expression. Thus, we can substitute values for parameters just as we would in any logical expression, and the result will be an instance of the original plan.

25. Goal Management

Goal Management concerns the mental processes involved in managing the lifecycle of one's goals. Goals are seen as entities that can be adopted, suspended, resumed, abandoned, violated, or achieved. A further distinction is made to identify adopted goals that are currently being pursued, that is, the focus of one's planning or execution capacities. People are also viewed as having multiple goals at any given time, with

prioritization among them that may be influenced through deliberative assessments.

Adding, Removing, And Modifying Goals

In previous chapters we have discussed the desires agents have in terms of their beliefs about the efficacy of certain actions for bringing about certain states; we have discussed the goals and subgoals in “The Plan” the agents are continuously executing as they go through the world in a purposeful fashion; and we have discussed the decisions that convert a causal belief into a subgoal. But in a way, this suggests too simple a picture. The Plan is a very dynamic structure, and the goals in it can be added, accomplished, abandoned, and modified. We define predicates for this aspect of planning in this section. This requires being able to talk about manipulating goals as independent entities and relating them to desires and to The Plan. In Chapter 31 we presented a very simplified picture of people as agents for whom a goal is either in or not in The Plan. But reality is much more complicated than this. Some goals are in The Plan and are actively being pursued at a given moment. Others have high priority, even though they are not currently being pursued. In the second section we characterize priority in terms of its determining factors, and then in the last section we talk about how priority influences which goals in The Plan are executed at a given time. All of this presents a much more nuanced picture of how beliefs and desires generate actions.

Priority

The order in which actions in a plan are carried out is usually underdetermined by the causal structure in the plan. A man may want to read a novel and go to the grocery store. There is no causal relation between these two actions that says one must be done before the other. Whether we execute an action at any given moment and the order in which we execute a set of actions are often determined by priority.

At least three factors play a role in determining the priority of an action, all of them scalar and two of them already explicated in previous chapters. The first is the importance of the action to the agent. This was introduced in Chapter 28 and constrained by its relation to the subgoal predicate – a goal is at least as important as its subgoals.

The more important an action is, the higher its priority.

The second factor is effort. We have not discussed this before. There is physical effort and mental effort. A commonsense theory of the human body and its activities would have to include an account of the intake and expenditure of energy, and physical effort would correlate with this. Mental effort is harder to characterize, but perhaps it could be explicated in terms of the time or number of mental operations required to integrate a proposition in the focus of attention into one’s belief system. A closely related concept which we have explicated is difficulty, in Chapter 15. The more obstacles one must overcome to achieve something, the more difficult it is. Difficulty and effort roughly correlate: Usually the more difficult something is, the more effort it requires.

We would like to say that, defeasibly, the more effort something takes, the lower its priority. We tend to do the short and easy jobs before the long and hard ones. Here we will translate that into difficulty. The more difficult an action, the lower its priority.

The third factor, not unrelated to the second is the likelihood of success. If I learn that I am bound to fail when I attempt an action, I will become less likely to attempt it. Thus, the more likely something is, the higher its priority, other things being equal.

To deal with the “other things being equal” clause, we have to make the simplifying assumption that importance, difficulty and likelihood are the only factors in determining priorities, and even with that we can only place very loose constraints on what it means for an action to have a priority.

Assessing And Prioritizing Goals

Goals are justified by an agent’s model of the world. A justification is a causal story relating the goal to some higher-level goal. The causal story is the product of an envisionment.

26. Execution Envisionment

One of the main purposes of envisioning is to imagine how action could change the future, or explain the past. Concepts of Execution Envisionment are fundamental across all envisioning processes, for example, explanation and expectation, but are distinguished in that each of them involves the envisioning of the causal implications of some action of an agent, whether they be one’s own or that of others. These concepts form the conceptual foundation for planning one’s own actions to achieve goals.

Executions And Their Envisionment

We plan. We act. We assess whether our actions have achieved our goals. If they haven’t, we replan. The expensive step in this process is the action. Fortunately there is a cheaper alternative. We can simply imagine carrying out our plan and assess the results of that. Its results are not as accurate as actually executing the plan, where we find out for sure what the results would be. But the cost of imagining is very low compared with the cost of acting, so the trade-off is worth it. We can also back-track when imagining, whereas backtracking from an action gone amiss is often a real problem.

Moreover, it is important for us to predict and explain the behavior of other entities in our environment. Among those entities are intentional agents. The best way to predict or explain their behavior is generally to assume they are executing some plan. If we surmise what the plan is and we envision their execution of that plan, then we are often able to predict or explain their actions.

We will discuss the actual executions of plans in Chapter 44 and subsequent chapters. Here we consider envisioning the execution of plans, viewed as a special case of envisioning a causal system. First we need two useful concepts.

Envisioning Success And Failure

An agent envisions success in an execution of a plan if the goal of the plan is one of the envisioned eventualities.

Specific And Arbitrary Contexts Of Execution

We talk about a “specific” entity as though this were a property. But in fact it is a relation between the entity and a set. The entity is a specific member of the set. But a third argument is required as well, and we must be explicit about it if we are going to characterize what “specific” means. An entity is a specific member of a set with respect to some purpose. We can say that this holds if and only if there is some property of the entity that does not hold for any of the other members of the set, and that property is causally involved in the purpose.

Envisioned Executions And Time

A moment in an envisioned execution is an instant in its time span. English expressions that make reference to moments in execution include “in the midst of” and “in the course of.”

27. Causes Of Failure

Try as we might, failure is common. Attributing blame to the root cause of failure is a central challenge in explanation. To simplify our efforts, we often resort to patterns of plan execution explanations that are widely applicable to everyday situations. Often in life, things don’t turn out the way we hope. Despite our best efforts to understand the world and plan a course of action that will achieve our goals, things go wrong somewhere along the way. When attempting to explain what went wrong, the explanations that people come up with are rarely very creative. It is much easier to rely on a certain pattern of explanation that is useful and satisfying across a wide range of failures than to understand really deeply the root of our problems. That is, it is easy to say that someone’s plan didn’t succeed because he or she didn’t have the skills necessary to execute it properly or claim that the plan would have been successful if only there were a little more time, rather than deconstruct the myriad of causal influences that were factors leading to the failure to achieve a goal.

These patterns of explanation for failed plans received a significant amount of attention among Artificial Intelligence researchers in the 1990s, largely emanating from the Yale University / Northwestern University computer science labs led by Roger Schank. Schank (1986) elaborated the notion of an Explanation Pattern, and pushed a multiyear, multidissertation effort to author a cognitive model of case-based explanation built on this idea (Domeshek, 1992; Jones, 1992; Ram, 1989; Kass, 1990; Leake, 1990; Owens, 1990), summarized in (Schank, Kass, and Riesbeck, 1994). Within the context of this work, a significant amount of effort was put into authoring comprehensive taxonomies of the types of explanation patterns that people used in understanding failed plans. Primarily, elements in these taxonomies were used as indices for the retrieval of stored explanations in support of a case-based reasoning process. The aim was to develop an indexing vocabulary (or content theory) of plan failure explanation patterns with broad coverage and utility for both scientific and engineering goals.

In this chapter we develop a new taxonomy of plan failure explanation patterns by combining the best of the taxonomies that have been developed in the past, synthesizing a taxonomy that is more comprehensive and elegant than any of the individual taxonomies that have been proposed in the past, and integrate it into the overall framework of this book. We close the chapter with a discussion of the utility of our taxonomy in scientific and engineering research.

Causal Complexes And Failure

When we try to achieve some goal state or event e , we find a causal complex s that we believe will result in e . We pick some element e_1 of s that does not hold and we try to achieve that, in hopes that it bring us closer to causing e to hold as well. Very broadly speaking, this process can fail in one of three ways:

1. We can be wrong in our belief that s is a causal complex for e .
2. We can execute e_1 but fail because other elements of s do not obtain.
3. Recursively, we can fail to achieve e_1 .

We can call these type 1, type 2, and type 3 failures.

To define these formally we need a strong notion of trying by some means, and a

notion of failure defined in terms of that. We will say that for agent a to try to achieve e by doing e_1 is for a to believe that e_1 is in a causal complex s for e and for a to do e_1 where a 's having the goal e causes a to do e_1 .

28. Plan Elements

Plan Elements are those concepts that further elaborate the internal structure of plans, enumerating their component parts. In addition to the concept of a subplan and its varieties, this area includes all of the programmatic control structure that defines a policy for how plans should be executed, including branch points, iterations, repetitions, and preconditions.

Suppose we are watching a workman doing his job. He looks around for his toolbox, and when he finds it, he opens it and takes out a hammer and nails. He pounds one nail after another into boards. Each step in this repetitive action is itself a repetitive action, hitting the nail with the hammer again and again until it is flush with the wood. He sweats, and from time to time as the sweat drips into his eyes, he takes out a handkerchief and wipes his brow. Five o'clock comes, and he puts his tools away and goes home. He comes back the next morning at nine o'clock for a day of similar tasks.

The workman has a goal he is trying to achieve, and he breaks this into subgoals that eventually bottom out in individual actions and bring him closer to the satisfaction of the goals. He needs to have certain tools and resources to do his job, and to have these, he needs to know where they are – resource preconditions and knowledge preconditions. He generally engages in repetitive actions because each repetition brings him a little closer to his goal. Situations arise and have to be dealt with immediately. To do his job, he has to be able to see, and when something interferes with this, he must somehow counteract it. His job is embedded in a larger structure of plans for his life as a whole, and this has to be aligned with periodic regularities of his physical and social environment. Thus, the fine structure of his actions emerges from his manipulations of the causal structure of the world as organized in his plans.

We as observers see individual actions happening one after the other. We make sense of them in part by recognizing similarities in successive actions and thereby recognizing repetitions. We see an action happening occasionally and realize it happens only when some condition arises. We thereby recognize conditional events. We see that some actions happen only at particular times or for particular durations. By these means we recognize the temporal structure of complex events, but we have not really interpreted his actions until we understand the causal structure implicit in them, that is, until we have recognized the plan he is executing.

Not every repetitive sequence of actions has a larger causal structure it fits into.

If someone is eating one piece of candy after another, he is probably not doing so in order to empty the bag or to fill his stomach. In fact, he may wish he had the strength to not take all the candy and not spoil his appetite. Each individual action is done for its own sake alone. But generally a “repeat until” structure has the goal of achieving the “until” condition, and a “while do” structure has the goal of negating the “while” condition. Each individual iteration may be supposed to cause the world to be a little closer to that goal state. Conditional actions are generally directed toward the goal of negating or exploiting the occurrence of the condition. By reasoning such as this, we infer the causal structure of events from their temporal structure.

29. Planning Modalities

Planning Modalities enumerate the different styles of reasoning that people employ in deciding what to do, broadly speaking. The most general concept is that of goal pursuit, an abstraction that includes deliberative plan construction, the adaptation of previous plans, the collaborative negotiation of plans with others, and several other planning modalities. Concepts related to three of these modalities, namely plan construction, plan adaptation, and design, are further elaborated in subsequent representational areas.

The Activity Of Planning

The process of achieving goals is in part a matter of using our causal knowledge to construct larger-scale causal knowledge. We then “decide” to use some bits of our causal knowledge, including this larger-scale causal knowledge, so they become part of The Plan we are actually executing. That is the source of our intentional behavior. We have called this whole process “pursuing goals,” represented with the predication (pursueGoal a g). The processes of constructing plans or larger-scale causal knowledge, making decisions about which of these plans to execute, and executing them are all parts of goal pursuit. Goal pursuit is our top-level intentional behavior. It is what we are doing when we are just “going about our business.”

Planning Activity And Other Agents

People exist in a world populated by other agents with other goals, and these goals can coincide, merely coexist, or contradict each other. We can define predicates for planning activity in the various cases.

Counterfactual Planning

Suppose Pat imagines how she would achieve world peace if she were president. In her plan there is a proposition (President Pat) which enables a large number of actions, and consequently is a subgoal in the plan. If Pat really is president, this is an ordinary plan and the plan bottoms out in this subgoal, because it already holds. If she is not, she may develop a plan to become president – declare her candidacy, begin campaigning, and so on. In this case also we have an ordinary plan. Being president is a subgoal and she constructs a plan to achieve the subgoal. But if the subgoal does not hold and she does no planning to achieve the subgoal, but goes on planning anyway to achieve world peace as though she were president, then she is engaged in counterfactual planning.

30. Planning Goals

Planning Goals concerns the goals that a person is trying to achieve in the process of deciding upon a plan of action. Here we are making a distinction between the types of goals that people are devising plans to achieve, and the preferences and priorities that people use to decide between candidate plans. Several of the concepts of planning goals concern the optimization of a resulting plan, such as the search for plans that take the least amount of time (minimize duration) or money (minimize value). Other planning goals prefer plans that include particular actions or events, or create opportunities or obstacles for other people. This area also includes three concepts for the goal of locating something (a person, place, or thing) that will play some other role in the plan.

Constraints And Preferences

As we develop plans for achieving specific goals, especially where many possible plans will work, we often seek to optimize various extraneous factors. For example, we may want to buy a sandwich but do so in a way that takes the least time. We may want to drive to a friend's house, but in a way that avoids traffic jams. We may plan a hike but leave open until we are there exactly where we will stop for our picnic lunch. This chapter focuses on these influences and modifications on plans, such as including and avoiding certain actions, enabling or blocking certain events, minimizing or maximizing certain values, locating relevant instances for general parameters, and maintaining the progress of the planning and the execution of the plan.

Including And Avoiding

One can prefer to include a particular action in a plan, as seen in the phrase “try to incorporate the step.” This is just the predicate `prefer0`.

One can prefer to avoid including a particular action in a plan, as seen in the phrases “try to avoid” and “only as a last resort.”

Enabling And Blocking

One can prefer to work toward plans that enable an event to occur, as in “set the stage for.”

Minimizing And Maximizing

One can prefer to try to minimize a value that is part of a plan, as illustrated in “less is more,” “be conservative,” and “frugal.”

Locating Instances For Plans

To instantiate an underspecified element in a plan is to add to the plan as a subgoal a property of that element that allows it to be identified. A context-dependent property of an entity that allows it to be identified is captured in the *wh*-words in English, for example, “I know who he is.” In Hobbs (1985) the predicate *wh* is used for this. Thus, to know the identity of something *x* is to know an *e* such that (*wh*’ *e x p*). The *wh* property is relative to a context, and in the planning realm, that can be specified by the plan *p*.

Maintaining Plan Progress

One can (and generally does) prefer to work toward plans that maintain the progress that has already been achieved. For example, if the top of a block is already clear and its being clear is necessary to achieve the goal, don't put something on top of it if you can help it. This is illustrated in the phrases “avoid backpedaling” and “keep moving forward.”

31. Plan Construction

Among the various ways that people decide what to do, Plan Construction is the process of devising plans from scratch. The process involves reasoning about the causal consequences of one's actions, and assembling a plan that would theoretically achieve the goal if correctly executed. This style of planning most closely aligns with the classical planning paradigm in AI, where operators are selected and ordered through

automated means–ends analysis.

The Planning Activity

In Chapter 36 we introduced the predication (planning e a g s), meaning that e is a planning activity by agent a to achieve goal g, where s is the sequence of plans that result from this activity. For example, the first element of s may be the limiting case of a plan, consisting of g alone, the second element might be that plus several immediate subgoals, and the last a complete, executable plan that should achieve the goal.

As agents go through the planning process, there is a range of moves they may use to develop their plans. At one end is instantiating an eventuality type in order to produce a plan more specifically tailored to the immediate situation.

Planning Process Control

A planning process is itself a plan-in-action, that is, a coherent sequence of actions that the agent carries out with the goal of coming up with a plan. Thus, it is subject to the same kinds of detailed characterization that other goal-directed behavior is. In particular, we can apply various aspectual predicates to it that we define in Chapter 45 on Execution Control.

Planning Subprocesses

In this section we define a number of moves an agent can make in developing a plan, from one plan in the plan sequence to the next. All of these definitions involve two successive plans p1 and p2 and say how p1 and p2 differ.

During planning, people can add detail to a plan in development, as seen in the expressions “flesh out the details” and “be specific about the plan.”

Planning Problems

An obstacle is something that causes a goal not to occur. Other words for this are “barrier,” “impediment,” and “impasse.” This definition includes both conditions as obstacles and entities whose properties are obstacles.

Selecting Among Candidate Plans

An agent may construct several possible plans to achieve a goal, and then must select among these candidate plans for a plan to execute. A candidate plan is a plan that the agent has in focus and is executable now with respect to a set c of constraints.

32. Plan Adaptation

Instead of dreaming up new plans to familiar problems, it is often easier just to do the same thing that worked last time. Things are not exactly as they were before, so it may be necessary to adapt to a changed set of circumstances. In the commonsense view of Plan Adaptation, people retrieve old plans that they had for previous situations and modify them in various ways to make them applicable to the current conditions. Sometimes these changes involve the addition or subtraction of plan steps, the modification of magnitudes or parameters, or the assignment of roles in multiperson plans. Some plans are straightforward to adapt, while others take a lot of mental work. In the end, people end up with plans that they like or plans that they reject.

Tweaking

One way people have for developing a plan to achieve a certain goal is to retrieve a known plan for achieving a related goal and tweaking it enough for it to achieve the desired goal. Suppose for example you go to a restaurant for the first time where you have to pay for your food before you eat rather than after; you retrieve your standard restaurant plan and tweak it by changing the order of events.

Tweaking occurs when an agent a takes a plan p_1 for achieving goal g_1 and forms a plan p_2 for a similar goal g_2 by replacing one step e_1 in p_1 with a similar step e_2 , yielding plan p_2 .

Types Of Plan Adaptation

We can think of a range of changes that fall under the heading of plan adaptation.

Plan p_1 may be our first attempt to achieve g_1 but it failed; g_2 is the same as g_1 ; and p_2 is the new plan with a modified step – e_2 for e_1 – in it. For example, p_1 might involve trying to pick up something with one hand, p_2 picking it up with two hands. Goal g_2 may involve nothing more than a change of time, so that only temporal parameters are involved in changing p_1 into p_2 .

More parameters may be changed. For example, p_1 can be the plan to go to a movie with a friend last Friday night, and p_2 is the plan to go to a different movie this Friday night at a different theatre, with a different friend, and with someone else driving.

More substantial changes may be involved. Suppose someone hears of a strategy to control nervousness during a choir performance by focusing on the conductor. He is nervous about giving a lecture to an august group of scholars. He retrieves this plan, modifies the activity, and decides to focus on a friendly face in the audience. Retrieved plans can be modified by changing values in a plan, as seen in the phrases “use more,” “do it with more,” “try increasing,” and “correct for.” This means that e_1 and e_2 are at relations of the same external entity v being at different points x_1 and x_2 in some system s .

Outcomes Of Plan Adaptation

Sometimes plan adaptation is worth the effort. Sometimes it is quicker just to develop a plan from scratch. We must have some measure of plan adaptation effort to be able to judge this. People do, as illustrated by the expressions “difficult to apply,” “complex application of,” and “straightforward modification.”

One measure of difficulty is the number of substitutions needed in plan p_1 to arrive at p_2 . By the theory of envisioning, a substitution is one step in the envisioning process. Adding an eventuality to a plan, as we do in generating a plan de novo, is also one step. Thus, this view of difficulty assumes each step in the envisioning process costs the same. Hence, this notion of cost is rather crude.

33. Design

Design is the planning of a physical configuration, of information to be communicated, or of an object intended to serve some function. The product of this reasoning need not include any actions to be executed, as when we talk about the successful completion of an architectural plan (e.g., the Burnham Plan for the city of Chicago). Instead, the goal is a satisfactory or optimal configuration of some composite entity that adheres to design constraints while supporting an intended use.

Artifacts

Consider a simple artifact – the coffee cup. It reflects a plan. We wish to transport coffee; that’s our highest-level goal. We do this by containing the coffee in a cup and then we transport the cup. One of the principal inferences about containment is that when the container moves, the substance that is contained also moves. Thus, transporting the cup transports the coffee. Similarly, when we attach a secondary entity to a primary entity, then moving the secondary entity causes the primary entity to move; this is one of the principal properties of attachment. Thus if we attach a handle to the cup and move the handle, the cup will also move, and so will the coffee. We can represent this plan for moving the coffee as follows:

Designs

A design is a pattern whose instances are artifacts, in the broad sense. We can design a teapot, a logo, a font, or a government. The two principal features of designs are that a design is intended to have instances and that a design and its instances are functional composite entities whose structure reflects a plan to realize the functionality. Thus, crucial to the design is the plan for the artifact. The plan behind the artifact is an instance of the plan for the design. In the same way that an artifact is a plan made concrete, a design is a specification of an artifact that concretizes a plan.

34. Decisions

A decision is the selection of a choice among options. The concepts listed in this area take the perspective that decisions should be viewed as a particular class of plan element, usually a part of a larger plan; for example, you’ll decide which way to go when we get there, rather than as a distinct planning modality. The process of decision making is compositional, involving the enumeration of a set of choices, deliberation over these choices, selection of the best choice, and (possibly) revision of one’s choice after the fact.

The Decision-Making Process

In Chapter 31 on Plans we introduced the predicate `decideTo`, which describes a transition of an action or other eventuality from something that is merely thought about or entertained to something that is part of the plan (`thePlan`) that the agent is actually executing. It expresses the conversion of thoughts into intentions. But in our thinking and in our discourse about decisions, we have a much richer model that we can appeal to concerning the processes involved in coming to a decision. We make decisions all the time and much of our mental life is wrapped up in how this is done.

Choices And Choice Sets

For any decision there are a number of possible choices (“range of candidates”). This is the third argument of the predicate `decision`.

The Process Of Deliberation

One way people come to their decisions is by predicting the outcome of a particular choice (“look before you leap”). We add a possible choice to our current world understanding and we envision from that to a set of consequences.

Justifications For Decisions

People have justifications for the choices that they select. A factor in a decision is a pro for the option selected or a con for an option not selected.

Consequences Of Decisions

There can be decision processes that are not carried to completion. When they are, however, they have consequences. By making a decision, people make things happen in the world.

35. Scheduling

Scheduling sits at the border between planning and execution, between dreaming and doing, between intention and action. The central concept in this area is that of a schedule, a mental agenda of the plans that one expects be executed at future points in time. The process of scheduling, consequently, is a deliberation concerning the time a plan should be executed.

Simultaneous Actions

Scheduling is a problem because we cannot do very many things at once. Preconditions for the actions must be met before the action is performed. Actions take a certain amount of time, whose normal duration we can estimate. There are physical and cognitive limits on the number of simultaneous actions an agent can perform at one time. All of these facts entail that time is a limited resource. Time places limitations and constraints on what can be done. Scheduling is figuring out how to work around temporal limitations and constraints. It is planning to satisfy these constraints. Gerald Ford purportedly could not walk and chew gum at the same time. We shouldn't talk while we are chewing, but we can interleave the activities and have a conversation over lunch. People cannot watch television, read the newspaper, and drive rapidly along a busy highway at the same time. It is difficult to do two unrelated actions with the same hand at the same time without substantial practice.

Schedules

A schedule is a mapping from a set of actions and other events into a set of temporal intervals or sequences. That is, it is a set of <event, interval> pairs. From a logical point of view, in a planning framework, we can view a schedule as a particular part of a plan, namely, those subgoals of the form (timeSpan t e) where t is an instant or interval and e is an action or other event in the plan. Thus, a schedule is an augmentation of a plan with temporal information. F

Scheduled And Pending Plans

Concerning scheduled and pending plans, there is range of possibilities. Do we want to say that a plan is scheduled when every event and nontemporal state is anchored to a specific real time, and that it is pending otherwise? Or do we want to say that a plan is pending when no event or nontemporal state is anchored to a specific real time, and that it is scheduled otherwise? Do we want to count as scheduled a plan whose left fringe subgoals are anchored to a specific time, but not necessarily other subgoals? Do we want to count it as anchoring when we only know the order of subgoals but not their specific times?

Schedule Preferences

In Chapter 32 on Goal Management we introduced the predication ($\text{preferGoal } a \text{ } g1 \text{ } g2$), meaning that agent a prefers to pursue goal $g1$ over pursuing goal $g2$. A schedule is a set and therefore a conjunction of the temporal subgoals of a plan. Thus, we can define a schedule preference in terms of goal preferences between schedules.

Scheduling Actions

For an agent to schedule a plan is for there to be a change from a state in which one has a pending plan to one in which one has a scheduled plan, where the subgoals of the pending plan are all included in the subgoals of the scheduled plan.

36. Monitoring

Plans often include the intentional triggering of some action based on some conditions of the situation. We imagine that stockbrokers would monitor the market for conditions that would prompt them to sell their portfolio. High school students might monitor the wall clock when taking a test to adjust the amount of time that they allot to answering each test question. People driving on an interstate highway might monitor the fuel gauge, the distance to exits, and their own bodies in making decisions about when to stop for gas at a service station. In the list that follows the term “monitor” is used both as a verb referring to the mental event of considering whether a situation matches some trigger conditions, and as a noun referring to a type of mental structure that describes these conditions and the actions that are to be triggered. This latter meaning of monitoring suggests that a stack or set of monitors might be active at any given time for a person, and that the adding and removing of monitors from this stack would be part of any moderately complex plan. It is also within this representational area that we can characterize the concept of waiting, both waiting for and waiting to, as a type of monitor whose conditions trigger the continuation of the remainder of a plan.

Monitoring Processes

As we go through our daily lives, we are executing plans and continually monitoring their effects to make sure they are actually achieving our goals and to make sure nothing else arises that demands greater attention. In a way, it is difficult to see how to capture this aspect of our experience in computational terms. It seems that we would need to wrap all action in a huge conditional that would check at every instant for threats and opportunities before executing the next step in the ongoing plan. But in fact the top-level loop in our behavior is a kind of “Interpret–Act” cycle. We look at the world and figure out what is going on, and then we do the next thing. Most of the time this requires little computation, since what is going on is what was going on an instant before, just advanced a little bit. When something unexpected presents itself, we must do more computation. It is not so much a matter of checking every possible opportunity or threat at every instant as it is of interpreting what presents itself, be it threat, opportunity, or the next expected thing.

Monitoring as we conceive it is a conscious subprocess of this “Interpret” stage of the loop. It is thoroughly integrated with the steps in the ongoing plans agents execute. It involves those threats and opportunities we are especially attuned to, given the nature of our plans.

We will say that an agent a monitors a set s of eventuality types. When a goal-relevant instance of any of these is perceived, this causes the agent to focus on that instance. This axiom is defeasible because sometimes we miss significant threats and

opportunities even though we are monitoring for them.

Characteristics Of Monitoring Processes

When monitoring, people are looking for monitor trigger conditions. These are the eventuality types in the set s .

37. Execution Modalities

There are several ways that people go about doing things. Different execution modalities are distinguished largely by the degree to which the action was planned ahead of time or with respect to temporal factors. In some cases, these modalities are necessarily exclusive, for example, an execution is unlikely to be both planned and spontaneous. In other cases, these modalities simply denote special concerns that an execution system would need to address, that is, the coordination needed for collaborative execution between multiple people.

Executions

A plan is a causal network that someone has in mind. Its actions may or may not actually happen in the real world. The plan may be designed to happen immediately or it may be designed with no intention of it ever happening. The plan, as a mental construct, is the same in either case. The two cases differ only in that in one the plan is executed, whereas in the other it isn't. For reasons like this it is useful to have as a distinct notion the "execution" of a plan. We will first define the execution of an action in a plan and then define the execution of a plan.

An agent a executes an action e in a plan p if and only if e is a subgoal of the plan, a is the agent of that action, and the fact that e is a subgoal in p plays a causal role in the occurrence of e .

Modalities Of Execution

Executions of plans can have a number of properties involving the agents' cognitive stance toward the executions. In this section we define a number of these modalities that our investigations have shown to be frequent.

In some cases, the event being executed is consciously planned by the agent.

Plan Execution And Time

Executions of plans are often composite actions that can be classified by the structure of the action that is executed. Our theory of event structure (Chapter 17) has provided us with the appropriate predicates for defining these categories.

Some executions are for iterative plans. That is, the agent is iterating through some set of entities, the notion captured by the predicate iteration.

38. Execution Control

Execution Control deals with the mechanisms of putting plans into action, and comprises two halves. First, Execution Control treats plans as sets of instructions and expectations about the effects the instructions will have on the world. It includes the concepts related to the launching and management of these instruction sets, including their starting, stopping, suspension, resumption, and continuation. Along with the basic control functions, this representation area includes concepts pertaining to the execution of particular types of plans with special specifications, including activities, performances, and skills – all of which may have plan representations that are not

comprised of explicitly executable sets of instructions.

The best laid plans of mice and men often go awry. The second half of the job of Execution Control is to notice when and where this happens, and to proceed only as long as they are viable. Included here are judgments of when conditions and preconditions in plans are satisfied by the current world state, whether deadlines in plans are met or missed, and the consideration of variables relevant to plans as they are being executed, for example, the elapsed time, the portion of the plan yet to execute, and the accumulated costs of resources that have been expended. This area also includes concept of an execution failure, the causes of which are the subject of a previous representational area, Causes of Failure.

Beginning Executions

In the previous chapter we defined the actions of executing a plan and executing a step in a plan, and we defined various concepts concerning external circumstances that might accompany such executions. In this chapter we will look at the internal structure of executions of plans, beginning with beginning.

The “left fringe” of a plan is the set of all those subgoals that can be done first, that is, those subgoals such that there is no subgoal that must be achieved before it.

Executions In Progress And Completed

The predicates `execute` and `executePlan` are defined in the abstract, without reference to specific times. At any given time t , an agent may be in the process of executing an action or a plan, and the agent may have completed an action or a plan. We will define time-dependent predicates using the present participle (“-ing”) for the former (`executing`, `executingPlan`) and the past participle “-ed”) for the latter (`executed`, `executedPlan`).

An agent is executing an action of a plan at time t if the execution takes place at that time.

Execution Costs And Outcomes

In Chapter 28 on Goals we developed a rudimentary theory of the cost of actions to an agent, in terms of what goals the action will force the agent to abandon. For example, if someone spends \$30,000 on a car, he will not be able to spend that money on other things he wants. The cost scale we characterized is consistent with costs being amounts of money, but it is not restricted to that. We defined the predication (`cost c e a`) to mean “ c is the cost of eventuality e to agent a . ”

The cost of an execution to an agent is easily expressed in terms of that predicate.

Abstract Plans And Their Instantiations

Plans can be at various levels of specificity. In a more abstract plan the identity of some of the participants and props may be unspecified and the times of some of the actions may not be fixed. Once we add properties to the plan, by elaborating the subgoals of the plan, then we are creating a new plan that is an instantiation of the old plan.

Plans And Aspect

The ways in which eventualities can unfold in time often go under the name of “aspect.” When we make aspectual statements about events, we look at the fine structure of the event as it is being executed and say something about that execution. The fine structure of events is captured in plans, or in an equivalent causal structure.

Therefore, this is the place in a theory of commonsense knowledge where a theory of aspect goes. We have already laid out the necessary infrastructure, and we can define various aspectual predicates in terms of that. Here we define a number of concepts that relate the execution of plans to the passage of time.

To start a plan is to start to execute an action in its left fringe.

Distraction

Distraction happens when there is a change for an agent from thinking of one thing x to thinking of another thing y , where some property of y (e.g., its proximity to the agent) plays a causal role in the change.

39. Repetitive Execution

Repetitive Execution enumerates the subset of Execution Control concepts involved in the iterative and repetitive execution of plans. Here distinction is made between repetition, which involves doing the same thing over and over again until some condition is met, and iteration, which involves applying some action to every element in some set of definite size. In programming language terms, there is some analogy to the flow-of-control distinction between a while-loop and a for-loop. Here a repetition or iteration in the singular refers to the execution of a subplan that is embedded inside the loop, affording the plural concepts of repetitions or iterations that are already completed or yet to complete. Here we also include the mental events needed for counting repetitions and iterations, affording people both the ability to discover quantity of sets in the world via counting as well as the ability to execute actions a definite number of times.

40. Mind–Body Interaction

Mind–body interaction concerns commonsense psychology at its periphery, where mind and body meet. In the commonsense view the mind and the body are distinct, yet have direct means of controlling and informing each other. The actions of the body, from moving one's hand to speaking some word, are seen as the result of some direction on the part of the mind. The directed action may fail or succeed, and in turn provides the mind with the feeling of its doing (body action perception), for example, as awkward or fluent, along with an array of other perceptions enabled by perceptual body parts, for example, one's eyes. These perceptions from the body share with the mind the resource that is one's focus of attention, with a scope that can be limited through concentration.

The English language is rich with words and phrases with meaning that is grounded in a commonsense theory of the interaction between the mind and the body. Concerning the mind's control of the body, we speak of motion that is dexterous, graceful, and nimble, or alternatively, awkward and clumsy. When the body moves without volition, we speak of spasms, reflexes, and stumbles. Concerning the perception of the world through the body to the mind, we speak of sensations and hallucinations. More subtly, we speak of the control that we have over our perceptions, as when we tune out distractions and pay attention to something. Conversely, we speak of the perception that we have of our own actions, when doing something feels unnatural or uncomfortable, or when we get the hang of doing something. When the normal functioning of the mind, the body, control, or perception is interrupted, we speak of being paralyzed, dazed, numb, asleep, knocked out, deaf, and tired, among a wide range of other mind–body states. In this chapter, our aim is to provide a theory that is

rich enough to capture subtle differences in the meaning of these and other concepts. Among the perils of formalizing a logical theory of mind–body interaction is the risk of trivializing the scholarship of generations of philosophers who have devoted their life’s work to this topic, specifically within the field of philosophy of mind. Chalmers (2002) reviews the rich history of philosophical debate on this topic, and outlines the various positions that different philosophers have taken. The classic dualism offered in Descartes’ *Meditations on First Philosophy* (2002) views a person as having both a physical body and a nonphysical mind. Descartes’ views are further characterized as interactionism, in that the mind and body interact in both directions. Huxley (1874) offers an alternative view of dualism, epiphenomenalism, where this interaction is only in the direction from the body to the mind. Ryle (1949) begins a philosophical shift away from dualism toward behaviorism, the view that the mind is not to be seen as something distinct from the body and steering it from inside, but as an aspect of the body’s own activities. The influence of neuroscience can be seen in the Identity Theory, put forth by Place (1956) and refined by Smart (1959), which holds that mental states are identical to their associated brain states: they are one and the same. Putnam (1973) argued against this view, noting that it is plausible that mental states are realizable by multiple brain states, and instead advanced the view of functionalism. Here mental states are seen as more abstract than their biological or mechanical realization, a view that has significant influence in contemporary artificial intelligence and cognitive science. Our aim, however, is not to encode a theory that reflects contemporary scientific or philosophical views of the nature of mind–body interaction, but rather to explicate the commonsense theory that nonscientists use to make everyday inferences. The result is a theory that most resembles Descartes’ dualism and interactionism, where a physical body and a nonphysical mind interact in both directions.

Mind And Body

Recall from Chapter 19 that we said a person has a body and a mind, and that the body has body parts. We specified that the body is a physical object. It seems most congruent with common sense to say that the mind is not a physical object. Arms, legs, heads, ears, tongues, vocal cords, and so on are body parts. A detailed explication of the body would be a part of a commonsense theory of anatomy and will not be done here. Body parts are capable of actions, many of them under voluntary control. Most body parts provide sensory information to the mind, including kinesthetic, pressure, and temperature information. Some body parts such as the eyes provide other specific kinds of information. Thus, we need to explicate the relation of the mind to bodily actions and to sensory information from body parts. We restrict our discussion to persons, rather than to agents more generally. So we do not include the wider range of nonhuman entities that are typically discussed in formalizations of agent-based theories, such as organizations and machines. Some of these agents may be seen as having minds and many certainly have various sensors and effectors. But all this would have to be axiomatized separately.

Awareness And Focusing

In our theory of memory in Chapter 23, we distinguished between the focus of attention and the memory. Concepts pass back and forth between the two all the time. We are conscious of what is in the focus of attention. It is sometimes convenient to distinguish a third, intermediate “region” of the mind, one’s awareness.

Perception

In Chapter 19 on Persons we introduced the predicate perceive and stated three of its properties: that what we perceive is external to the mind; that an entity's being near a person enables perception of the entity; and that intact sense organs enable perception.

Bodily Actions

A bodily action by a person is a movement of one or more of the person's body parts. There is a range in the voluntary character of bodily actions. If a car hits a man and he goes flying through the air, his body parts are moving from one place to another, so under our current definition it would be a bodily action. But all the causality is provided by the external world.

In a reflex action, such as a kick resulting from a tap on the knee or blinking caused by a sudden flash of light, the external world provides some of the causality and the body provides the rest, but the conscious mind is not involved. It is directly caused by a stimulation from the environment, that is, by the motion of some entity (a hammer, a beam of light) from somewhere outside a body part to the body part, where some property of the body part enables the bodily action. (This last condition rules out being hit by a car as a reflex.) But the action is not willed.

Perceiving Bodily Actions

Among the phenomena we can perceive are some of our own bodily actions. Our kinesthetic sense enables awareness of the position and movement of parts of the body by means of proprioceptors in the muscles and joints. Certainly there are some bodily actions we are not aware of, such as the normal working of our digestive systems or our heart, or the circulation of the blood in our cardiovascular system. We will say that a bodily action defeasibly causes a person to perceive it.

Controlling Perception

Among the body parts that can be moved in a voluntary action are the sense organs. We can direct our gaze and focus our eyes onto a particular object in the environment, or we can look away from it. We can touch something to see how it feels, or retract our hand if it feels slimy. We can cup our ears to hear something better, or change the orientation of our head to determine the direction of a sound better, or plug up our ears with our fingers to avoid hearing the sound. We can flare our nostrils to smell something better or hold our noses to not smell it. The purpose of all these actions is to perceive something we would not otherwise perceive or to avoid perceiving something we would otherwise perceive. We can call these actions "sense organ actions."

Levels Of Capability And Activity

In Chapter 28 on Goals we defined what it is for a functional composite entity to be intact, namely, for each of its functional components to be able to execute its function. Such an entity is "impaired" when some of its functional components are not able to achieve their functionality.

States Of Consciousness

The theory of mind-body interaction presented here postulates that a person has both a mind and a body, which are connected by the two channels of sensation and control. The mind, the body, the sensation channel, and the control channel each have

a three-valued state, namely, they are active, impaired, or inactive, or in the case of the body, intact, damaged, or destroyed. The normal sensation channel operates to translate the stimulation provided by the world to beliefs in the mind. The normal control channel translates the will to do an action into the performance of the action in the world.

We can elaborate our theory with an account of action and perception that is dependent on the status of the mind, the body, control, and sensation. Under this account, the mind's perception of the world is a consequence of the world stimulating the body, and is influenced by the channel of sensation. With an intact body, an active mind, and active sensation, this perception yields beliefs; that is, seeing is believing. Likewise, the body's actions are a consequence of the mind's will, influenced by the channel of control. With an active mind, an intact body, and active control, will yields action.

States of consciousness are bundles of inferences we can draw about the control a person's mind has over his or her body and the influence perception can have on the mind. That is, a state of consciousness is a quality we hypothesize of a person that then allows us to draw many inferences about how he or she is likely to behave. Some states of consciousness, such as being drunk, have identifiable causes, while others, such as sleep and dreaming, normally do not.

When somebody is "fine," his or her body, mind, and sensation and control channels are all intact or active.

41. Observation Of Plan Executions

Observation of Execution relates to a special case of Plan Following, where a person is following a plan that is being executed by another person. In the general case, the ability to follow the actions of another person is a critical component of collaborative execution, where two or more people attempt to execute a shared plan.

However, many of the concepts in this representational area deal specifically with two more specific situations. First is the observation of another person's execution of some performance when the plan is known, for example, for the purpose of assessing their abilities. For example, an Olympic judge follows the performance of a gymnast as they perform their routine, and assesses how well they adhere to some idealized performance specification. Second is the observation of another person's execution when their plan is unknown, for example, for the purpose of learning how to do something. For example, novice surgeons will observe the performance of many surgeries during their medical residency in order to induce the complex plans that they will later execute themselves.

Observing The Plan Executions Of Other Agents

In the previous several chapters we have addressed issues involving agents executing plans to achieve goals and monitoring these executions to make sure they do advance toward the goal. We have also gone deeper into the possible states of an agent's perception of the world. Among the events an agent will perceive are the efforts of other agents to achieve their goals, and that is the focus of this chapter. This will take us into issues of instruction, performance, skill, and evaluation. This topic lies at the edge of commonsense microsociology, as interactions among agents crucially involve their observations of each other's purposeful actions.

People can observe some aspects of the execution of plans by other agents.

Instructions

In a previously developed theory of information structure (Hobbs and Mulkar-Mehta, 2013), communication is characterized as the presentation by one agent (the speaker) of a symbol x to another agent (the hearer), causing the second agent to think of a corresponding concept y , where the mapping from x to y is a result of conventionalized rules in a social group s that both agents belong to. This is encapsulated in the predication

Performances And Their Specification

A performance is the execution of an explicit plan before an audience.

Skill

Ideally when we attempt to carry out a plan, we do it the right way. The “way” a goal is achieved is simply by executing the subgoals and having that cause to goal to come about, that is, by executing the plan.

Evaluation

To evaluate a performance is to come to a belief about where that performance lies on a skill scale for such performances, as a result of witnessing the performance.

42. Emotions

Emotions play a special role in the commonsense theory of the mind. Indeed, many might not view emotions as part of the mind at all. One might say, “Don’t listen to your mind; listen to your heart,” or “Don’t think too much about it; trust your feelings.” Likewise, it is common for people to ascribe emotional reasoning to a spiritual component of their existence, distinct from their body and mind. This perceived dualism between emotional and nonemotional reasoning likely contributed to the widespread popular acceptance of the idea of emotional intelligence during the 1990s, with only tenuous links to psychological research (although see Salovey and Grewal, 2005). However, high-level explicit theories of the role of emotion in reasoning seem to play little or no role in characterizing the lexicon of emotion-related terms in the English language, which is largely dominated to references to specific emotional states.

The use of emotion terms has been studied extensively within the field of linguistics, with particular attention to cross-cultural differences. Good introductions to this line of cross-cultural linguistic research can be found in edited volumes by Athanasiadou and Tabakowska (1998) and Harkins and Wierzbicka (2001), and in the monograph of Wierzbicka (1999). Within computational linguistics there has been recent interest in creating large-scale text corpora where expressions of emotion and other private states are annotated (Wiebe, Wilson, and Cardie, 2005).

In our own analysis of language related to concepts of emotion, we were particularly struck by two characteristics of emotion vocabulary that distinguish them from those of other representational areas. First is the sheer quantity of single words that reference emotion states in the English language; there are hundreds of words that English-speakers use to describe how they are feeling. Second is the low level of polysemy within this set. There are lots of emotion words, and they are unambiguous. In the vocabulary list that follows, we also include feelings and other sensations,

and cluster emotion concepts by how they relate to other representational areas of commonsense psychology.

To paraphrase the Dick and Jane primers of an older generation, we see, think, and do. Intelligent agents perceive the environment around them, they consider how that environment will impact what they seek to achieve and modify their plans accordingly, and they execute actions that will achieve their goals in this environment. Their actions alter the environment, and possibly their own and others' beliefs and goals, and the cycle begins again. This is the basic cycle of cognition.

Emotion can be viewed similarly. We see, feel, and do. One view of emotion is that it is leftover reptilian cognition. It is a system of cognition that enables rapid and usually appropriate responses to threats and opportunities. In any case, it is best to view emotion, not as it is sometimes viewed, as opposed to or an alternative to cognition, but as a special variety of cognition.

Emotions, like cognition, mediate what happens to us and how we respond to it, but generally in a much more direct way than cognition. We can analyze the internal structure of emotions to some extent, in terms of the accompanying cognitive elaborations and in terms of levels of arousal. But for the most part the important facts about emotions involve their relations to the outside world. Specifically, to understand the role of emotion in human life, one has to know what kinds of things cause various emotions, and what kinds of things are caused by various emotions. We are all able to fake emotions, and the way we fake them is by knowing and using these two categories of causal facts.

A number of different theories of emotions have been proposed in the psychological literature (e.g. Scherer, 2010). One of the most popular is the so-called "appraisal" theory. Appraisal is the combination of "see" and "think"/"feel." We perceive our environment and we judge how it relates to our beliefs and goals, in order to operate in this environment in the optimal way. The patterns of appraisal in appraisal theories are the specific causal rules associated with different emotions, telling us what classes of event cause which emotions.

The patterns of appraisal, or their associated emotions, trigger specific coping strategies, or physiological, behavioral, or cognitive reactions. In our terms, these links are the rules about what classes of actions by the person are caused by the emotions. Emotions exist because these responses usually have adaptive value.

"Dimensional" theories of emotion posit two or more dimensions along which affective states can be characterized. In a dimensional theory, the external environment causes the person to be in a particular state, fully characterized by an n-tuple of the values along each of the dimensions. This state then induces various behaviors. The problem with dimensional theories is that in mapping the environmental stimulus into such an n-tuple, the specificity of response is lost and must be recovered by cognitively assessing the cause of the particular state. For example, you see a bear in your path as you are hiking through the woods. You feel a generalized fear state, rather than specifically being afraid of the bear. Then you have to assess your situation to figure out what it was that triggered your fear state and thus, what an appropriate response would be. Such an account may in fact be true. It may be true that being afraid of the bear is an after-the-fact interpretation of a subconsciously produced state of generalized fear. But this is decidedly not the commonsense view of how emotion operates. Moreover, it would be counter to the whole approach of this book, based on first-order logic, to arbitrarily lose track of arguments of predicates in the middle of a psychological process. Dimensional theories are fundamentally propositional – FEAR rather than (fear I bear).

Nevertheless, the dimensions that dimension theorists propose are aspects of emotions that we need to take into account. Three commonly proposed dimensions are pleasure or valence, arousal or intensity, and dominance. We specify for the emotions we characterize whether or not they are varieties of happiness or unhappiness, as our way of capturing pleasure or valence. We discuss and partially characterize the scalar notion of intensity. Dominance concerns the person's ability to deal with the environmental stimulus, and can thus be characterized in terms of the response that particular emotions trigger.

"Anatomic" theories of emotion focus on the physiological changes that are triggered by stimuli, rather than actions that result from deliberation. We do not deal with these aspects of emotion, as we have not explicated a commonsense theory of anatomy and in any case commonsense approaches to emotion deal with physiological effects only at a very coarse level, for example, an association of anger with being hot.

"Communicative" theories of emotion view emotions as states that we display for the purpose of communicating beliefs and likely actions to others, thereby facilitating social interaction. This perspective is especially relevant to computational work on emotion in human-computer interaction, where the whole point of emotional displays by the computer is to elicit particular social responses. Our view of this is that of course people exploit displays of emotion to convey things to or infer things about other people. People can exploit virtually anything they perceive or experience for communicative purposes. But this is parasitic on an emotion system that evolved for independent reasons. By knowing what class of actions is caused by a particular emotion, as we explicate in our axioms, observers of a person's action can work backwards from the action to the emotional state, and from there back to possible causes or forward to other possible actions. The person has thereby communicated all this information to the observer, whether intentionally or unintentionally.

We can identify five events involved in a specific emotional response:

1. Some external (or internal) event or trigger occurs.
2. It is perceived and evaluated with respect to the person's possibility of attaining his or her goals, for example, survival.
3. There is a subjective feeling, for example, anger, fear, or happiness.
4. There are various physiological manifestations, for example, a smile or grimace, dilation or contraction of the pupils, and dilation or contraction of the blood vessels.
5. The person takes some external (or internal) action in response to the emotion, for example, fight or flight.

In 1 and 5 we normally think of the event and action as external, but it could be internal. Suddenly thinking of an old friend who died can trigger sadness, and the response to an emotion might be a change of belief.

William James (1890, Chapter 25) argues, contra a naive view that 3 causes 4, that 3 in fact just is 4. That is, 3 is the way we experience subjectively the physiological effects in 4, in the same way that "tired" is the way we experience a high concentration of lactic acid in our muscles. It's just two different levels of description. They can be viewed as identical, or we can say that there is a kind of "how to" or "how did" causal relation between them – 4 is the way 3 is "implemented."

In any case, we do not need to concern ourselves with this issue, as we are not explicating the physiological accompaniments of emotion.

In addition to the mediating role emotions play in causal knowledge, all emotions have an associated level of intensity. This interacts with the causal facts. In general, the more important the goals involved, the more intense the emotion. The more intense the emotion, the greater the effect of the responses. Because intensity applies

to all emotions, we will discuss this first.

It has been argued that there are a small number of basic emotions. A common proposal identifies five – happiness, sadness, anger, fear, and disgust. We will consider these first in the list of emotions we discuss. The first two are relatively long-term moods that result from one's general success or failure and result in characteristic patterns of behavior. Happiness is caused by one's goals being satisfied, and promotes further action. Sadness is caused by one's goals not being satisfied, and suppresses further action.

Then as a way of demonstrating the nuances of meaning that we can capture in the framework we have built up, we examine a number of synonyms and near synonyms of "happy."

The other three basic emotions – what we call "the raw emotions" – are specific responses to specific events. These are discussed next.

Many other emotions, if not all, can be seen as cognitive elaborations on these basic five. Thus, resentment is a kind of anger. We next axiomatize the emotions characterized by Ortony, Clore, and Collins (1990), and relate them to the basic emotions, insofar as appropriate.

Intensity And Arousal

Intensity of emotion is a scalar notion that is probably impossible to define precisely. But we can delimit its interpretation by means of two of its properties, both defeasible, one relating to what causes the emotion and one relating to what the emotion causes.

The first property is that, defeasibly, the more important the goal involved in the emotion, the more intense the emotion. Fearing that your life is in danger is usually more intense than fearing that you might stub your toe.

In Chapter 28 we explicated a more important relation between eventualities in terms of an agent's goal hierarchy – supergoals are more important than subgoals. Toward the end of this chapter we label various emotional states as emotional states, by the predicate `emotionalState`, which takes an eventuality and a person as its arguments.

Happiness And Sadness

Specific emotions can in part be characterized by what abstract classes of eventualities cause them. Happiness is normally caused by the belief that one's goals are being satisfied.

Shades Of Happiness

We have rich and nuanced emotional lives. We can be glad that something happened without being cheerful about it. We can be elated without being euphoric. We can be happy because we feel gratitude, gloating, vindication, or pride. In this section we give characterizations of several synonyms and near synonyms of "happy," as a way of demonstrating the scope of the theories we have developed.

We characterized happiness in terms of its causes (goals being satisfied) and its effects (higher level of activity). Sometimes in discourse the cause of the happiness is focused on.

The Raw Emotions

What we are calling the raw emotions are ones that are generally in anybody's list of the basic emotions. There are three. The raw emotions are fear, anger, and disgust.

Unlike happiness and sadness, they are specific responses to specific threats, and the actions they trigger have immediate survival value. They are deeply biological. For sure, reptiles do not feel such complex emotions as vindication or resentment. But we can easily imagine reptiles feeling something like fear, anger, and disgust. We can give a very coarse-grained, primitive, “reptilian” characterization of the raw emotions in the form of a schema. Threats are of two kinds – the kind that eats you and the kind that you eat that then kills you from inside. Disgust is the response to things that you eat. It triggers efforts to eject the threat. Anger and fear are responses to things that eat you, and correspond to the fight-or-flight responses. In general, if the threat is something you can defeat or cope with, you get angry and fight. You attempt to move it to a place where it is no longer a threat or to destroy its capacity to do you harm. If it is something that can defeat you or that you can’t cope with, you become afraid and flee. You cause yourself to move to where it is no longer a threat. Of course, these emotions extend beyond culinary concerns. As with all emotions, we can characterize abstract situations in which they arise, and abstract responses we have to those situations.

The person is embedded in an environment. Something in the environment poses a threat, that is, there is the possibility that one or more of the person’s goals, such as survival, is going to be defeated. The first question is where the threat is located. If it is in the external environment the emotion triggered will be fear or anger. If it is literally or metaphorically “inside” the person, it will trigger disgust. The response to disgust is to eject the threat. The response to fear is to remove one’s self from the threatening environment. The response to anger is to remain in the environment and to modify it so that the threat no longer exists.

Envy And Jealousy

Envy and jealousy are closely related and are often used interchangeably. Envy is displeasure resulting from seeing the success of others, and is very much related to the particular success. Jealousy seems to be more directed toward the person.

Suppose an athlete about to receive an Olympic gold medal is told that he must share it with a rival. He will not feel envious, because his rival has nothing that he doesn’t have. But he may well feel jealous.

One possible characterization of envy and jealousy is that in envy we feel resentment over a nonexclusive goal that the other person has and we don’t, whereas in jealousy, we feel resentment over a mutually exclusive goal that someone else has. We are more likely to label our feelings jealousy when someone has an exclusive relationship with someone we secretly admire. Our feelings toward someone who is very wealthy is more likely to be described as envy.

Liking And Disliking

As Heider (1958, p. 140) points out, liking and disliking are not so much emotions as dispositions to feel particular emotions in specific circumstances.

We like a state or event if it makes us happy, defeasibly of course.

Emotional States And Tendencies

It is sometimes convenient to talk about emotional states in general, independent of their particular content. Insofar as the five purportedly basic emotions cover the field, we can characterize emotional states by five axioms.

Appraisal And Coping

People's emotions are generally caused by their appraisal of the situation, that is, by their coming to believe something about the environment. An emotional appraisal happens when a change in belief causes an emotional state.